AI Adoption in America: Who, What, and Where

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Abstract

We study the early adoption and diffusion of five AI-related technologies (automated-guided vehicles, machine learning, machine vision, natural language processing, and voice recognition) as documented in the 2018 Annual Business Survey of 850,000 firms across the United States. We find that fewer than 6% of firms used any of the AI-related technologies we measure, though most very large firms reported at least some AI use. Weighted by employment, average adoption was just over 18%. Among dynamic young firms, AI use was highest alongside more educated, more-experienced, and younger owners, including owners motivated by bringing new ideas to market or helping the community. AI adoption was also more common in startups displaying indicators of high-growth entrepreneurship, such as venture capital funding, recent innovation, and growth-oriented business strategies. Adoption was far from evenly spread across America: a handful of "superstar" cities and emerging technology hubs led startups' use of AI. These patterns of early AI use foreshadow economic and social impacts far beyond its limited initial diffusion, with the possibility of a growing "AI divide" if early patterns persist.

* Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. This study relies on confidential micro-data available to approved projects through the U.S. Federal Statistical Research Data Centers. All results have been reviewed to ensure that no confidential information is disclosed. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data (Project No. P-7508509, Disclosure Review Board (DRB) approval number: CBDRB-FY20-095, CBDRB-FY20-331, CBDRB-FY21-041, CBDRB-FY22-074. CBDRB-FY22-246. CBDRB-FY23-0309). We thank the Editor, two anonymous referees, Tim Bresnahan, John Eltinge, Nathan Goldschlag, Rob Seamans, and participants in the Columbia/Wharton Management, Analytics, and Data Conference and Georgetown's McDonough School of Business strategy seminar for excellent comments andfeedback. Financial support from the Stanford Digital Economy Lab, the Ewing Marion Kauffman Foundation and the Social Sciences and Humanities Research Council (SSHRC) of Canada is gratefully acknowledged. All errors remain our own.

1 Introduction

Artificial intelligence (AI) has the potential to spark a revolution in business and the economy (Spulber 2011b). However, systematic evidence on who the using firms are, what types of business they are in, or where they are located, has been slow to accumulate. This makes it difficult to understand AI's economic and managerial implications or ground predictions in objective data. Yet, the consequences of uninformed AI investment and policy may be far-reaching (Agrawal et al. 2019). To address the evidence gap, we leverage new large-scale survey data to characterize early AI use across the U.S. economy, surfacing factors likely to shape its impact on workers, firms, and society.

Organizational contexts have long been credited with shaping the outcomes of technology adoption, including recent advances such as data analytics (Wu et al. 2019; Brynjolfsson et al. 2021). Today, concern is growing over the contexts and consequences of advanced technology use. Income inequality is on the rise, with growing links to digitization (Autor et al. 2020; Tambe et al. 2020; Barth et al. 2023). AI-related technologies, in particular, are attracting attention and debate (Brynjolfsson 2022; Acemoglu and Restrepo 2020). Meanwhile, limited insight into organizational factors associated with the use of AI limits guidance regarding specific areas of promise or concern.

To shed light on patterns of AI use among firms, we rely on a new nationally representative survey, the Annual Business Survey (ABS), conducted jointly between the U.S. Census Bureau and the National Center for Science and Engineering Statistics (NCSES) within the National Science Foundation. The ABS solicits information on firm-level use of several advanced technologies—including those currently associated with progress in AI—over the prior year, beginning in 2018. The initial wave of the survey was sent to a large, representative sample of over 850,000 firms in the private non-farm economy, with a high response rate of 69%.[1] We link this to the Longitudinal Business Database (LBD), which provides administrative data on employment and revenue throughout the firm life cycle for a near-universe of U.S. firms. This allows us to characterize early AI use in a sample of 474,000 firms which, once weighted appropriately, represents over 4 million firms nationwide. We further leverage rich founder and organizational data for a subsample of 75,000 startups to provide insight into the relationship between AI use and firm dynamics.

We first analyze *who* adopted AI, defined as firm use of at least one AI-related technology[2] in

production. Our representative statistics indicate that just under 6% of firms nationwide used AI as of 2017. Yet most very large firms (over 5,000 employees), reported at least some AI use. Intensity of use varied from merely testing (1.1%) to using AI in more than a quarter of production (2.2%). More-intensive use was more-prevalent among very large firms (nearly 30% used AI intensively). Manufacturing and information were leading sectors, with adoption rates of roughly 12%, each.

Our ability to observe *what* was adopted, in terms of specific business technologies (including, but not limited to, AI), reveals key interdependencies. Firms using AI were more likely to rely extensively on both digitized information and cloud computing, suggesting an important role for other "enabling" technologies (Gambardella et al. 2021; Goldfarb et al. 2023) in AI adoption. Also, while typically studied in isolation, we find combined use of AI and robotics: most firms using robotics in production also used AI. In addition to having important implications for physical automation in key sectors, such clustering of emerging technologies has, historically, indicated mutually-reinforcing innovation and broader economic impact, overall (e.g., Rosenberg 1963).

Dynamics matter, as well. Visibility to the firm life cycle reveals that, conditional on size and industry, younger firms were somewhat more likely to report using AI. Given the central role of young firms in innovation and economic dynamism (Spulber 2011a; Acemoglu et al. 2018; Haltiwanger et al. 2013; Decker et al. 2017, 2020), we expect that patterns of AI adoption among growth- and innovation-oriented startups will be essential to understanding the rate and direction of AI use in the economy, moving forward.

We thus delve deeper into a large subsample of roughly 75,000 startups (5 years or younger) for which we have unusually detailed data on owner characteristics and motivations, startup financing, and innovation and business strategies. Across a range of specifications, we find that AI-using startups tended to have younger, yet more highly–educated and more-experienced, leaders. Founders motivated to bring new ideas to market and help the community were more prevalent among AI-users than those pursuing so-called "lifestyle" entrepreneurship (e.g., emphasizing flexible hours or work–family balance). Key markers of high-growth entrepreneurship, specifically venture capital (VC) funding, high initial capitalization, and reliance on formal intellectual property (IP) protection, also predict AI presence. Recent innovation in products or processes were further associated with startups' use of AI, as were growth-oriented business strategies.

We argue that these differences—many determined at founding (Guzman and Stern 2015, 2016)

or suggesting distinct strategic commitments (Ghemawat 1991; McElheran et al. 2019; Li 2023)—are fundamental to understanding AI's trajectory in the broader economy. Further, while markers of high growth potential (Azoulay et al. 2020; Guzman and Stern 2020) are predictive of AI use, when we condition on direct measures of such factors, a significant association persists between AI use and revenue growth—including later revenue gains. While establishing causality is beyond the reach of our data, this pattern points to important linkages between AI use and firm performance.

Finally, while the spread of AI use across the country remains in its initial phase, the potential for an "AI divide" across regions and cities is attracting concern. Investigating *where* early AI adoption was located, we find considerable concentration among AI-using startups. The expected influences of California's Silicon Valley and Bay Area are apparent. Yet the share of startups using AI was also high in less-obvious places, such as the metro areas surrounding Nashville, San Antonio, Las Vegas, New Orleans, San Diego, and Tampa. Once we weight by employment to account for worker exposure to AI, Riverside, Louisville, Columbus, Austin, and Atlanta also show concentrated AI use. This distinction between firm adoption and employee exposure may be critical to assessing the "future of work," and where it may shift, soonest, for individuals across these geographies. Observing this difference is a key advantage of our research setting, and draws attention to a broader set of locales than typically associated with AI uptake.

This rich snapshot of early AI use constitutes the most extensive and detailed evidence on AI adoption, to date. Our main objective is to document—rather than explain—a number of new stylized facts. Taken together, they point to numerous organizational and geographic intangibles that have already begun to shape AI use in ways that have been unobserved, far less appreciated.

Yet, these patterns of early use signpost where AI is likely to have the most (or at least the earliest-observable) impact. For example, the strong connection between process innovation and early use of AI in production points to a likely hurdle—one observed with prior waves of technological innovation (e.g., Bresnahan and Greenstein 1996; McElheran 2015) and commonly associated with "general purpose technologies" (GPTs) (Bresnahan and Trajtenberg 1995)—to broader diffusion of the technology. Drivers of process innovation likely play an under-appreciated role in promoting AI use and economic impact. Yet the separate importance of formal intellectual property protection suggests that early adopters have invested in protecting their innovative efforts. This has important implications for how firms intend to compete using AI, as well as the likelihood

that AI-related innovation will be observable in patent and related data, moving forward. More broadly, higher AI use among firms with prosocial founding motivations may provide guardrails against unethical or biased applications (Cowgill and Tucker 2019), while early concentration among large incumbents may augment existing "superstar" firm dynamics (Autor et al. 2020; Tambe et al. 2020; Camuffo et al. 2022). Geographic concentration evokes similar concerns about an AI-fueled "digital divide," though in some perhaps less-expected places.

Much work remains to unpack these and other implications of our findings. They illustrate, however, numerous ways in which future assessments of AI's economic and social impacts may hinge on choices over input and outcome measures, data availability, and which subpopulations contribute meaningfully to underlying data-generating processes.

This study makes a few key contributions. First, our approach allows us to characterize the early—and, by extension, the most likely later—AI adopters in the largest and most representative sample of firms to date. This approach is critical for getting the "denominator" right in estimating rates of AI diffusion and identifying likely areas of impact. Prior work has measured AI diffusion and worker exposure using O*NET task descriptions (e.g., Brynjolfsson et al. 2018; Felten et al. 2021; Eloundou et al. 2023), online job postings (Alekseeva et al. 2021; Acemoglu et al. 2022; Babina et al. 2022; Bessen et al. 2021; Goldfarb et al. 2023), and patents (Webb 2019; Hong et al. 2023; Miric et al. 2023). However, direct firm-level measures of AI use are rare, as are rich organizational covariates—particularly at scale. Privately held, younger, and smaller firms—essential to contextualizing the phenomenon in the overall economy—are also typically lacking. The hurdles producing such measurement gaps are non-trivial (McElheran 2018; Miric et al. 2023), yet not insurmountable—as we show here. This new survey may serve as a guide for follow-on work across other country and research contexts.[3]

Second, by tying AI use to owner characteristics and motivations in a large subsample of young firms, we add to the nascent literature on entrepreneurial motivation (e.g., Cassar and Meier 2018; Gans et al. 2019; Shah et al. 2019; Guzman et al. 2020; Ganguli et al. 2021). We document novel linkages between startup leadership, high-growth entrepreneurship, and frontier technology use.

Third, our findings contribute updated insights to an important literature on drivers of information technology (IT) adoption, including organizational complements to IT use (Ichniowski et al. 1995; Brynjolfsson and Hitt 2000; Bresnahan et al. 2002; Bloom et al. 2012, *inter alia*). Recent

work in this vein has emphasized innovation and business strategy as complements to data and analytics (Wu et al. 2020; Brynjolfsson et al. 2021). Identifying the correlates of AI adoption is an essential step in this agenda. Further, our focus on elusive high-growth startups, as well as the large number of typically-unobserved organizational details we explore, add new dimensions to this ever-expanding and timely body of work.

Finally, this study speaks to a literature on the geography of innovation and technology diffusion, dating back to Griliches (1957), that has gained momentum with the rise of digital technologies (e.g., Forman et al. 2012; Tambe 2014; Kerr and Robert-Nicoud 2020; Bloom et al. 2021). While certain familiar patterns of geographic concentration emerge with respect to AI, we find nuances in the relationship between urbanization and AI adoption. For instance, being located in a large urban area is associated with a greater likelihood of a startup using AI, yet this disappears once we control directly for characteristics that typically go hand-in-hand with urbanization, such as serial entrepreneurship, access to startup funding, and successful innovation. Moreover, we emphasize that employment-weighted geographic trends are indicative of AI's impact on workers, not solely on firms—a nuance that leans heavily on our comprehensive administrative data.

Our analysis shows that early AI use was low, skewed, and closely linked to the high-growth entrepreneurship essential to economic dynamism. However, its prevalence in the largest firms and key geographies raises concern about the sustainability of past dynamics. Our findings surface a number of novel insights into the industrial, technological, and organizational context of AI use to inform how we navigate AI's pitfalls and potential, moving forward.

2 Motivation and Prior Work

Our framework for exploring the prevalence and implications of AI use builds on Zolas et al. (2020), which reported on the use of advanced business technologies, more broadly. Here, we focus specifically on AI and the technological and organizational context surrounding its use.

2.1 Technology Diffusion and Its Implications

A longstanding literature seeks to understand patterns of technology diffusion and their economic and social outcomes (e.g., Griliches 1957). A key motivation is the relationship between technology use and productivity (e.g., David 1990; Brynjolfsson and Hitt 1996), as well as the role

of "general-purpose technologies" (GPTs) in innovation and technical progress (Bresnahan and Trajtenberg 1995). Recent digital advances are gaining attention, including "big data," analytics, and data-driven decision making (Brynjolfsson and McElheran 2016, 2019; Tambe 2014; Wu et al. 2019; Brynjolfsson et al. 2021). Yet, the speed of technological change raises questions about the applicability of prior intuitions when it comes to AI (e.g., Agrawal et al. 2019; Goldfarb et al. 2023).

Recent research into AI has measured its prevalence based on O*NET task descriptions (Brynjolfsson et al. 2018; Felten et al. 2021; Eloundou et al. 2023), online job descriptions (Alekseeva et al. 2021; Acemoglu et al. 2022; Babina et al. 2022; Goldfarb et al. 2023), or combining the latter with patent and/or research publication data (Webb 2019; Bessen et al. 2021; Hong et al. 2023).

Direct firm-level measures of AI use are rare, particularly outside of Europe (where samples tend to be smaller—see, e.g., Czarnitzki et al. 2023; Hoffreumon et al. 2023). Studies of AI in other countries, such as Canada and China, have had to rely on yet different approaches to measurement (Alexopoulos and Cohen 2018; Lu et al. 2023; Beraja et al. 2023). Privately held, younger, and smaller firms—essential to contextualizing the phenomenon in many economies and capturing key dynamics—are systematically underrepresented, as are their organizational details. Our study makes a contribution in terms of the scale of the data set, its representation of the diversity of U.S. firms, and the richness of organizational details (particularly for startups, discussed below).

Earlier IT diffusion research has established the importance of commonly observed firm characteristics such as industry, size, and location (e.g., Forman et al. 2005). We characterize AI use along these familiar dimensions, while disentangling often-confounded ones such as age and technological sophistication. The former is difficult to disentangle from size, despite being central to firm life cycle dynamics (Kueng et al. 2014; Haltiwanger et al. 2013). Our exploration into the latter, which focuses on data availability and cloud use, is motivated by recent attention to "enabling" technologies (Rathje and Katila 2021; Gambardella et al. 2021; Kapoor and Teece 2021), including recent findings linking data, cloud computing, and AI (Goldfarb et al. 2023; Lu et al. 2023), particularly among startups (Bessen et al. 2022). Studying AI in isolation from these other pervasive innovations could obscure important interdependencies shaping its diffusion and impact, particularly in our heterogeneous, nation-wide sample of firms.

Similarly, while AI and robotics are frequently discussed together as automation technologies primed to transform the global workforce (Raj and Seamans 2019), they are typically studied in

isolation. Robots primarily automate physical tasks (Dixon et al. 2021), whereas AI automates cognitive ones (Agrawal et al. 2019). Yet robotics is poised to increasingly rely on AI for greater autonomy and reprogrammability. Understanding their co-occurrence may be key to understanding how AI is mobilized in production, with attendant firm and labor-market outcomes.

2.2 Entrepreneurship and Technology Use

A key advantage of our research setting is detailed organizational data for a large sample of non-public, primarily young and small firms. Younger firms, in particular, are often missing from standard data sets, despite being vital to economic dynamism and innovation (Spulber 2011a; Chava et al. 2013; Decker et al. 2017, 2020; Haltiwanger et al. 2013, 2016; Acemoglu et al. 2018; Guzman and Stern 2020). We examine AI use among startups in detail, focusing first on characteristics that are largely fixed at founding and, as we establish empirically, associated with revenue growth early in the firm life cycle. Our overarching motivation for exploring the relationships to follow is that, if early AI use is concentrated among high-potential firms, we might expect greater economic and social impacts of AI as these firms scale. Conversely, if AI-related economic gains remain elusive (Brynjolfsson et al. 2021), our findings may point to key frictions or measurement challenges for future study. Finally, if we can identify a relationship between AI use and later firm growth, this may indicate where causal links between AI use and firm performance will ultimately be found.

2.2.1 Owner Characteristics: Education, Experience, and Age

We leverage a growing body of evidence that certain early organizational markers can indicate a firm's growth potential and reliance on innovation in its strategy (e.g., Guzman and Stern 2020; Botelho et al. 2021). Such high-growth entrepreneurship has further been linked to a number of founder characteristics and motivations (Levine and Rubinstein 2018; Gans et al. 2019; Guzman et al. 2020). Intuitively, founders with advanced degrees are likely to lead firms that are more technology-focused or managed using more-advanced technology. An entrepreneur's prior experience predicts firm survival and performance (Nanda and Sørensen 2010). Serial entrepreneurship, in particular, may be correlated with a better assessment and exploitation of the business opportunities advanced technologies offer, or reflect the adaptability of an entrepreneur to new technology (Lafontaine and Shaw 2016). Though unexplored in prior work, we anticipate that greater owner education and experience will be associated with greater AI use, all else equal.

With respect to age, Azoulay et al. (2020) find the mean founder age for the fastest growing new ventures to be 45, which is older than commonly understood — particularly in technology-focused sectors. Yet it is consistent with tension between "vintage-specific human capital" (younger workers have better facility with more-recent technology) and experience (or other intangible skills that take time to accumulate). When the latter complements digitization, this creates the best "fit" (and associated compensation) for mid-career workers at IT-intensive firms compared to *both* older and younger employees (Barth et al. 2023). However, prior work has not explored how this tension nets out in the context of AI adoption, whose newness may require even more familiarity with the most-recent technology, particularly at the top of the organization.

2.2.2 Founder Motivation

Next, we expect that aspirations of entrepreneurs will shape the rate and direction of AI's impact, at least among startups. The smaller older firms that lag in advanced technology use (Zolas et al. 2020) may have been founded to pursue objectives other than growth, lacking the use cases, complementary inputs, and scale to justify AI adoption. The literature has delineated "lifestyle" from "high-growth" entrepreneurship (e.g., Hurst and Pugsley 2011). We anticipate that lifestyle-focused owners will be less likely to make the technology and organizational investments required to deploy AI in a meaningful way.

Recent work also distinguishes between founders motivated by prosocial aspirations and/or workplace values and those motivated primarily by earning potential (Cassar and Meier 2018; Guzman et al. 2020; Shah et al. 2019; Ganguli et al. 2021). While we cannot observe specific applications of AI, we expect founder motivation may foreshadow the potential for ethical or prosocial applications, versus ones with more-adverse consequences (e.g., Cowgill and Tucker 2019).

2.2.3 Startup Financing

Firm trajectories have further been linked to aspects of startup financing. Selection by—and synergies between—VCs and entrepreneurs have been linked to later growth.[4] A relationship between venture capital, high-growth entrepreneurship, and AI use may therefore arise for a variety of reasons. One channel could be selection: VCs may be able to better identify startups capable of developing or leveraging new AI technologies. Another could be treatment: the type of products and processes venture capitalists encourage may rely more significantly on AI use.

Higher initial capitalization may also reflect private information about firm growth potential. We thus also explore its relationship to AI use. Care is required, however, as capital intensity, even within narrowly defined industries, may reflect production strategy (McElheran et al. 2019) as much as firm "quality," and is positively associated with the presence of other advanced technologies in manufacturing plants (Dinlersoz and Wolf 2018). We leverage direct insights into dimensions of innovation and business strategy (rarely observed at scale) to disentangle source and level of financing from growth or "quality" as factors in AI adoption.

2.3 Startup Innovation and Business Strategies

Many of the organizational factors discussed above are relatively fixed early in the firm life cycle. However, we are also interested in the relationship between AI use and startups' innovation and business strategies, recognizing that they may be co-determined. On this front, we are motivated by a large and influential literature on complementarities between organizational features and technology use (e.g., Ichniowski et al. 1995; Brynjolfsson and Hitt 2000; Bresnahan et al. 2002; Bartel et al. 2007; Bloom et al. 2012).[5] This literature often abstracts away from causality among complements, or else argues that organizational characteristics tend to be quasi-fixed compared to IT (e.g., Tambe et al. 2012). This literature has further focused predominantly on large incumbents and has only begun to explore more-recent innovations such as "big data" and analytics (Aral et al. 2012; Wu et al. 2019, 2020; Brynjolfsson et al. 2021). Thus, the extent to which prior patterns hold with respect to AI use is poorly understood, particularly among dynamic young firms.

In addition, the notion that strategic commitments (Ghemawat 1991) to certain innovation and business strategies might complement or promote technology use (rather than vice versa) has received scant attention in economics and management strategy. We contribute new insights, here. To begin, process versus product innovation is an important distinction—indeed, a tension—in firm dynamics and innovation strategy (Cohen and Klepper 1996). Whether and how this distinction matters with respect to frontier IT use, however, is less clear. On the one hand, informal process innovation may be critical for successful adoption of any new technology (e.g., Bresnahan and Greenstein 1996; McElheran 2015). On the other, formal intellectual property (IP) protection may signal commitments to innovation and future growth (Guzman and Stern 2020; Dinlersoz et al. 2021, 2023) that may promote AI use. Understanding nuances in this relationship may be essential

to unpacking adoption and growth dynamics related to AI.

Finally, recent work indicates that dimensions of a firm's business strategy may influence new technology uses, complementary management practices, and how technology performs across different organizational contexts (McElheran et al. 2019; Wu et al. 2020; Brynjolfsson et al. 2021; Li 2023; McElheran and Jin 2023). We extend this line of inquiry to understand AI use among startups with different strategic objectives.

2.4 Geography of Startups

Research into both technology diffusion and entrepreneurship has long emphasized the importance of geography for understanding patterns of technology use, location choice, and economic outcomes (e.g. Forman et al. 2012, 2016; Delgado et al. 2010; Alcácer and Chung 2014). Recent work has pointed to growing concentration, both of new digital technologies (Tambe 2014; Bloom et al. 2021), and of startup activity (Guzman and Stern 2015). VC- and innovation-related spillovers may further contribute to the emergence of dominant hubs for startups leveraging specific technologies (Jaffe et al. 1993; Samila and Sorenson 2010), though this is often difficult to disentangle from other agglomeration economies (Kerr and Robert-Nicoud 2020). While AI is still in a relatively early stage of its geographic diffusion (Hong et al. 2023), initial technology hubs tend to retain a considerable share of the relevant employment (Bloom et al. 2021). Thus, AI's future path will likely depend significantly on its early geographic dispersion. While we cannot observe shifts over time, uncovering geographic patterns in AI adoption, especially among startups, is therefore a key objective of our analysis.

3 Data and New Descriptive "Facts"

This study provides the first in-depth look at AI adoption from the Annual Business Survey (ABS). Introduced in 2018, the ABS consolidates three earlier data collections capturing important features of U.S. firms. New designed-for-purpose questions on technology use were added to existing questions that provide, for relevant firms, ownership and owner characteristics, startup financing, intellectual property use, and other details of innovation and business strategy.[6] We describe data construction and novel descriptive statistics here, beginning with our baseline sample. For additional details, see Data Appendix B.

3.1 Baseline Sample

The 2018 ABS was sent to 850,000 firms nationwide. Approximately 583,000 firms responded to the survey, 573,000 of which were linked to the Longitudinal Business Database (LBD). The LBD is curated by Census to provide a comprehensive panel of microdata on employment, revenue, and payroll for the private non-farm economy (Chow et al. 2021). The LBD tracks firms from their birth as an employer to death, accounting for mergers and acquisitions. This provides the basis for our systematic exploration of early- and later-stage growth patterns for firms in the ABS.

Leveraging the LBD mitigates concerns that we could overlook dynamic entrepreneurial firms, which are often excluded from standard data sets comprised of larger incumbents (e.g., Compustat). Conversely, it ensures we do not omit older firms that might be systematically excluded if sampling depended on the presence of advanced technology (such as participating on a digital platform). We call this sample of 573,000 firms the ABS–LBD linked sample (column 1, Table 1).

For most of our analyses, we restrict attention to the 447,000 firms for which we have sufficiently complete information on AI use. Descriptions of average firm size and age for this baseline sample are presented in column 2 of Table 1. We weight the firms in our samples to better represent the firm size, age, and industry distributions of the population of employer businesses in the U.S.-wide Business Dynamics Statistics (BDS).[7] The vast majority of U.S. firms are very small,[8] which is reflected in our baseline sample. The average firm, here, has roughly 62 employees, but only 21 when weighted. Average firm age is around 16 years (regardless of weighting). Weighting makes our sample representative of over 4 million firms across the U.S. economy (column 2, Table 1).

3.2 New Measures for the Digital Age: Digitization, Cloud Computing, and Advanced Business Technologies

The technology module in the 2018 ABS contains three new and connected questions on digitization and technology use. The first queries firms' reliance on digital information (data), widely regarded as a key input and even a prerequisite to more advanced uses of digital technologies (Brynjolfsson and McElheran 2016, 2019). A second oft-cited enabler is sufficient computing power to store and analyze massive quantities of data. Thus, the second question gathers data on the extent to which firms rely on cloud services, which have shifted the cost structure and speed of access to

IT resources for many firms (Jin and McElheran 2017; Goldfarb et al. 2023; Lu et al. 2023).

The third question, and the primary focus of this paper, asks about the use of various advanced business technologies in producing goods or services, including intensity of use. Specific definitions are provided in Table 2 and include five commonly associated with AI (bolded): machine learning, machine vision, natural language processing, voice recognition software, and automated guided vehicles.[9] Tabulated responses to these five items, conditional on some use (keeping in mind that only 2% of firms in the overall unweighted sample reported any use of AI in production or testing), are reported in Figure 1. Note the considerable dispersion across both specific technologies and intensities of use. Adoption rates were below 3%, across the board, ranging from a high of 2.9% in the case of machine learning, to a low of 0.8% for automated guided vehicles. Testing of each technology—presumably a leading indicator of use in production—was even lower (less than 1% each), suggesting that these levels were not poised to change dramatically in the wake of our study.

3.3 Early AI Use: Low and Skewed

For our main analyses, we collapse different technologies and intensities of use into a single binary indicator of *any* use of AI. According to this definition, the share of firms using at least one AI-related technology as of 2017 was 5.8%. The share of firms testing at least one of these technologies (but not yet using it in production) was 1.1% (Figure 1).

This level of AI diffusion is significantly lower than that reported elsewhere, such as the European Commission survey of AI (Kazakova et al. 2020) or other private surveys by McKinsey (Chui and Malhotra 2018), Deloitte (Deloitte 2018), or PwC (PwC 2019). However, these surveys do not claim to be representative, oversampling larger, often publicly traded companies. In contrast, our baseline sample includes the many small firms for which AI adoption is quite limited—yet whose inclusion allows us to estimate the correct "denominator" for country-wide AI usage rates.

While these low average rates are informative—and useful for counteracting potentially excessive "exuberance" related to AI—they paint an incomplete and potentially understated picture of AI use in America. As for many advanced business technologies (e.g., Zolas et al. 2020), early AI use appears highly skewed and concentrated among larger firms. Figure 2a reports AI use by firm size for different intensities.[10] Note that the majority of firms with more than 5,000 employees reported at least some use of AI. Over a quarter of the largest firms reported using AI intensively.

This skewness in both AI presence and intensity underscores the potential impact of AI not only on firms, but also on employees at these firms. The concentration of AI in the right tail of the firm size distribution means that worker exposure to AI may be considerably higher. Returning to Figure 1, employment weighting (in parentheses) suggests machine learning prevalence of 8.5%, and employment-weighted usage of *any* AI technology of roughly 18.2%. While we explore employment weighting to understand other "stylized facts," below, we focus first on firm-weighted statistics.

3.4 *Who* Adopts AI: Industry, Size, and Age

Our rich administrative data grants insights into key characteristics of early AI users—the *who* of interest in this study. Figure 2b reports AI use and intensity by sector. Manufacturing- and information-sector firms led in early AI adoption, with roughly 12% extensive-margin adoption rates, each. Both also led in intensity of use, with health-related firms close behind. Lagging sectors included construction and retail trade, at roughly 4%, each. Finance, insurance, and real estate ("FIRE") sectors had adoption rates below 6% with limited intensity, despite ranking highly in digitization of information—see, e.g., Figure 2 in Zolas et al. (2020).

Table 3 takes a closer look at the non-manufacturing industries (at the 4-digit NAICS level) where AI was most prevalent. On a firm-weighted basis, medical and diagnostic laboratories had a striking uptake above 23%. Other high-AI industries tended to be "high tech," such as software publishing (nearly 16%) and computer systems design (14%), though with exceptions. Data processing and related services, which includes cloud services, also reported AI use around 14%.

We control for this industry-based heterogeneity at the 6-digit NAICS level (a degree of granularity difficult to achieve in standard data sets) to uncover other patterns of interest.[11] Table 4 reports the likelihood of using AI by size and age, including industry-by-state controls. Conditional correlations show AI usage increasing monotonically with size. In column 1, firms in the top percentile of their industry's employment distribution were nearly 9 percentage points (pp) more likely to use AI compared to below-median firms. In contrast, we find AI presence decreasing almost monotonically with age, though magnitudes are smaller. The oldest firms were 1–2pp less likely to use AI than firms younger than their industry's median age (Table 4).[12]

3.5 *What* Gets Adopted: Technological Details and Interdependencies

Column 2 of Table 4 sheds new light on *what* technologies get adopted, adding indicators for digitization and cloud computing. Controlling for the above observables, firms that had also come to rely on digital information were more likely to adopt AI. In particular, firms that reported having at least one type of information in digital format (over 65% of the ABS-LBD linked sample) were 2pp more likely to report using an AI-intensive technology. An even larger correlation is evident with cloud computing: firms that reported purchasing cloud services for at least one IT function (43% of the overall linked sample) were 5pp more likely to use an AI-intensive technology. Indeed, these margins of technology use appear further interconnected, as can be seen in Figure 3, which shows a Sankey Diagram of firms that used one or more of each type of technology. Firms that reported early AI use were most likely to perform their data analysis using cloud-based IT services. These same firms were also most likely to store their production information digitally (Figure 3).

We further observe variation by industry in terms of *what specific AI-related technology* (among those in Table 2) was most prevalent. As shown in Table A1, while more-generic "machine learning" was highly prevalent in computer systems and data processing industries, industries that are not always considered "high-tech" led in the use of specific technologies. For example, automated guided vehicles (AGV), while used by fewer than 1 percent of firms nationwide, were present at roughly 9 times that rate in support activities for crop production in the agriculture sector. Similarly, machine vision was most prevalent in technical and trade schools in the education sector (nearly 5%), while voice recognition was most prevalent in healthcare settings and legal services.

This industry-based heterogeneity also hints at potential complementarities between AI and other labor-saving, capital-intensive technologies such as robotics. Robotics adoption based on ABS responses (1.3%) was significantly lower than that for AI. However, the overlap with AI appears fairly significant, with the majority of robotics adopters (57.5%) reporting using at least one AI technology. Figure 4 provides a sectoral breakdown of the overlap of firms that use AI, robotics, or both, with the greatest overlap in the Manufacturing, Wholesale, and Education sectors.[13]

3.6 AI in Entrepreneurship: Owner Characteristics, Startup Conditions, and Firm Strategies in AI-Using Startups

A key advantage of the ABS is that, in addition to new technology measures, it contains detailed information related to the owners of the firm, their reasons for owning the business, financing at startup or acquisition, intellectual property strategy, and related innovation and business strategies—key issues in entrepreneurship research.[14] Thus, given the importance of young firms in job creation and innovation, the role of age in predicting AI adoption, and such unusual visibility to the organizational context surrounding AI use, we narrow our focus to analyze, in depth, the roughly 75,000 startups (0–5 years old as of 2017) in our sample. See Table 1, Column 4.

3.6.1 Owner Education, Experience, Age, and Aspirations

We focus on three main dimensions of the firm context that are typically unobservable in large samples: owner characteristics and aspirations, startup conditions (specifically, financing level and source), and firm strategies related to innovation and growth. The ABS asks approximately 20 questions about up to four business owners; we focus on the primary owner (i.e., the owner with the largest share) in Table 5.

Starting with education, we construct an indicator of having a Master's, Doctoral, or Professional degree. While 22.4% of all primary owners in our startup sample reported holding such an advanced degree, this rises to 31.5% among AI-using startups (row 1, columns 1 and 2).

With respect to experience, the important variation is well-captured by an indicator for any prior ownership (in row 2). Owners of AI-using startups were relatively more likely to have owned a prior business (43.8%) compared to the full startup sample (41.4%).

On an unconditional basis, owners of AI-using starts tended to be younger. For instance, in row 6 of Table 5, only 24.7% of such owners were 55 years or older (Column 2), compared to 26.6% of the overall startup sample (Column 1). That said, for ages in the range bracketing "peak entrepreneurship" (Azoulay et al. 2020), averages were closer (row 5). We examine this in more detail in a regression framework, below.

For owner (primarily founder) aspirations, when comparing various motivations for owning the business between owners of AI-using versus other startups, the reasons with the biggest differences between the two groups were wanting to "Help and/or become more involved in my community"

(difference of 6.6 in Column 4) and "Best avenue for my ideas/goods/services" (difference of 5.1) The share of owners in AI-using firms reporting these reasons are 30.3% and 61.8%, respectively, compared to 24.1% and 56.9% of all startup owners in our sample (Table 5). These growth-oriented and prosocial reasons show a sharp difference—and are relevant with respect to prior research—when compared to "lifestyle reasons" for entrepreneurship. For the analyses to follow, we rely on an indicator which receives a value of 1 if the primary owner reports either "flexible hours" or "balance work and family" as a "very important" reason for owning the business. These reasons were very important for the owners of 52.8% and 55.1% of AI-using firms, compared to 54.8% and 56.4% of all primary owners in the startup sample.

3.6.2 Startup Financing

Next, we explore source and size of capitalization of the firm at startup or acquisition. Focusing on venture capital, we find a sizable unconditional difference for AI-using startups, with 2.9% reporting being funded by venture capital compared to only 0.9% of all firms in the startup sample. AI-using startups, on average, also reported larger values of startup capital: 46.5% reported startup capital of more than $25,000 compared to only 41.4% of all startups (Table 5).

3.6.3 Innovation and Business Strategies

The ABS asks directly about process and product innovation, separately, across the prior 3 years (2015–2017). Process innovation was more prevalent among AI-using startups, at 39.3%, versus 19.9% across the startup sample. Product innovation, even more so: AI-using startups reported product innovation at a rate of 66.3%, while the startup average was 53.6% (Table 5).

AI-using startups relied on formal IP to protect innovations at higher rates, as well. The prevalence of patents owned or pending among such firms was 5.2%, versus 2.1%, on average. IP was reported to be "very important" at a rate of 49.4% among AI-users, while it was only 19.5% among the entire startup sample (Table 5).

Finally, we construct a measure of how "growth-oriented" the firm's business strategy is, based on whether it reported that introducing new goods or services, reaching new customer groups, or opening up new domestic or export markets were "very important." AI-using startups espoused a growth-oriented strategy at a rate of 77.5%, compared to an average rate of 79.5% (Table 5).

3.7 *Where*: The Geography of AI-Using Startups

The last set of analyses focuses on *where*, turning to the geographic distribution of AI use across the U.S. Looking at Core-Based Statistical Areas (CBSAs), we identify a set of "hubs" for AI use across the country.[15] We focus on young single-unit firms, as they are straightforward to situate geographically compared to larger, multi-unit entities. Note, however, that single-unit firms accounted for approximately 97% of firms and 75% of establishments, respectively, according to the U.S. Census Bureau's Business Dynamics Statistics for 2017. Thus, while our main analysis retains the focus on startups from above, we leverage the unusually good coverage of our data to map all AI-using single-unit firms, regardless of age, in the appendix (see Figures 6a and 6b).

We assess AI prevalence within each CBSA in two ways: first, as the share of single-unit startups that use AI; and, second, in terms of the share of employment among single-unit startups accounted for by AI-using firms.[16] The ranking for the largest CBSAs is shown on a map in Figure 5a.[17] Bubble sizes represent the number of AI-using startups present in the CBSA, while the color gradient indicates the percentile rank of the CBSA in terms of the AI usage rate—lighter colors correspond to greater prevalence of AI use among young single-unit firms in that CBSA.

Some high-level patterns are apparent. Areas known for pioneering technologies, such as Silicon Valley in California and Research Triangle in North Carolina,[18] show high concentrations of AI use among startups as shown by the lighter colors. Yet "hubs" for AI use are apparent throughout the South and the West, as well. The share of startups using AI was high in the metro areas surrounding Nashville, San Antonio, Las Vegas, New Orleans, San Diego, and Tampa.

The Northeast and Midwest display lower AI *intensity* as share of the firm population (note darker colors). However, larger metro areas typically have a larger number of firms in general, including AI-using single-unit startups. Thus, large metro areas such as New York City, Los Angeles, Chicago, and Washington D.C. display prominently in terms of bubble size (i.e., have many single-unit startups that use AI), while *within* these areas AI usage is not particularly intense.[19]

The picture changes to some extent when we consider employment-weighted adoption rates across CBSAs, as shown in Figure 5b. This provides a view of worker exposure to AI. Additional locations attain "hub" status using this approach, primarily in the Midwest and Mid-Atlantic, including Louisville, Columbus, Austin, and Atlanta. Locations in the South tended to fall, in

a relative sense, due to employment weighting. These shifts can be attributed to a variety of factors. For instance, certain locations have a few young single-unit AI users that are relatively large compared to others in the same location, whereas in other locations AI-using single-unit firms may be numerous yet too small to make up a large share of employment. Differences in industrial composition across locations also matter. For instance, manufacturing is a leading AI-using industry (Figure 2b). Thus, large AI-using manufacturing startups clustered in the Midwest receive greater weight in this analysis, despite comprising a small fraction of their CBSA's total single-unit startups.

Table A2 provides a list of CBSAs, ordered alphabetically by name, alongside their share of single-unit startups with AI use and overall rank along this dimension among others in the list. Firm-weighted rankings (corresponding to Figure 5a) are in Column 1, while Column 2 relies on employment weighting (Figure 5b). Overall shares for micropolitan, small-, and medium-sized CBSAs are included as well. While AI usage rates are generally lower in the smaller CBSA groups, they are not too far below the corresponding rates in the largest CBSAs. We return to the role of CBSA size in AI adoption in Section 4.3, particularly once we condition on key advantages of large urban locations for startups, below.

4 Regression Analysis

4.1 Methodology

Moving beyond novel descriptive statistics, our empirical approach assesses whether the "fixed at founding" organizational features available in the 2018 ABS may help predict patterns of AI adoption, as well as how the unconditional patterns described above might shift in a more richly specified regression framework. This will help identify the relative importance of various drivers of AI use, as well as clarify the extent to which it is associated with growth-potential early in life. The latter is central to understanding the potential impact of AI in the broader economy.

To do this, we proceed in steps, first establishing—via ordinary least squares (OLS) regression—the extent to which these traits do, indeed, identify startups that turned out to be "high quality" in terms of being high-growth early in their life cycles. Because we also seek to understand the role of more time-varying organizational complements, we include in our assessment innovation and

business strategies that may (a) be correlated with early revenue trajectories, and (b) also predict AI usage (see Section 2). For these analyses, we measure early growth trajectories by taking the average of the annual log-difference revenue (r_t) growth rate over the first three years of a firm

$$g = \frac{1}{3}\sum_{t=1}^{3} \ln(\frac{r_t}{r_{t-1}}). \tag{1}$$

After using this measure to assess which firm characteristics are associated with early firm growth, we fit a linear probability model using OLS to explore the extent to which the same firm features may also predict AI use.

Beyond characterizing patterns of early AI use, we are interested in the extent to which the AI use we observe may be related to firm performance. To explore this connection, we examine whether AI presence had a significant association with firm growth after controlling for the rich firm characteristics we observe. To do this, we further measure average growth over the three years 2016–2018 (later-stage growth) around the survey reference year of 2017, again using (1). Note that the later-stage growth is only used for the owner and revenue sample of firms (column 3, Table 1), as startups are by definition in early stages of their life cycle (the first 5 years after birth).

Our analysis does not aim to establish a causal link between firm characteristics and adoption, nor between adoption and firm growth—a challenge left for future work. We also note that the sample selection criteria underlying our analyses limit the applicability of our findings to the entire firm population without additional caveats. In particular, our measure of firm growth is based on revenue, and because revenue data in the LBD goes only as far back as 1997, our analyses with respect to growth are limited to our owner and revenue sample—a subset of ABS respondents that were born in or after 1997. Dropping firms born before 1997 excludes many firms that tend to be larger as a function of being in business longer.[20] Moreover, our owner and revenue sample only contains the firms which report having at least one owner with a nonzero share of the firm. Therefore, firms that did not report any ownership information, which disproportionately includes publicly owned firms and firms concentrated in specific sectors (such as manufacturing), are also excluded from our more detailed analyses. To the extent that these excluded firms tend to be large, we would expect them to be more likely to use AI, based on the patterns we observe in the rest of the data. In addition, we are limited to those firms that survived through 2017 and responded to

the survey in 2018; differential survival rates of adopters versus non-adopters prior to 2017 could further introduce biases whose sign is difficult to predict.[21]

4.2 Markers of High-Growth Entrepreneurship

We begin by establishing conditional correlations between early revenue growth in startups and the factors discussed in Section 2. We regress the early average revenue growth rate (equation 1) on our three broad groups of firm characteristics—first on each individual group, and then on all groups together—controlling for state-by-industry and firm age. While never statistically significant, we also include controls for owner gender in all specifications.

Column 1 of Table 6a reports the estimated coefficients for owner characteristics. An owner holding an advanced degree or previously owning a business is associated with a higher early growth rate. Specifically, a business with an owner who both holds an advanced degree and previously owned a business is associated with a 4.4 percentage-point (pp) higher early growth rate. With respect to age, we observe a monotonic relationship: startups with an owner aged 35 or older tend to have a lower early growth rate than those with an owner younger than 35 (and firms with owners 55 or older tend to exhibit even less growth than those with owners aged 35–54). We also observe that having a "lifestyle" reason as being very important for starting the business has a negative (though not statistically significant in this model) point estimate.

Column 2 of Table 6a explores startup conditions. VC investment and larger initial capitalization are both associated with higher startup growth. For example, a startup funded by venture capital and with a startup capital amount of more than $1 million has a 30pp higher early growth rate than one not funded by venture capital and with less than $25,000 in startup capital.

Column 3 reports the coefficients for innovation and business strategies. Each of process and product innovation, intellectual property use, and growth-oriented business strategy is conditionally associated with higher early growth. This is particularly so for patent ownership (7.7pp higher early growth rate) and having a growth-oriented innovation strategy (5.4pp higher rate).

Column 4 includes all of the owner and firm characteristics together, showing the robustness of the patterns in Columns 1–3. In fact, the indicator of "lifestyle" reason as being "very important" for starting the business not only remains negative but also increases in magnitude and becomes statistically significant. That is, controlling for owner characteristics, startup conditions,

and innovation/business strategies, an owner who reported a "lifestyle" reason as being "very important" for owning the business is associated with a 0.8pp lower early growth rate. In Column 5 we show this same specification replacing the "lifestyle" reason indicator with indicators for all possible reasons included in the ABS, which are reported separately in Table 6b. We find that the coefficients on both "Flexible hours" and "Balance work and family" (i.e., the reasons that make up our "lifestyle" indicator), as well as on "Unable to find employment," are negative and statistically significant (Column 1 of Table 6b). We also find positive and statistically significant coefficients on "Best avenue for ideas/goods/services," "Help and/or become more involved in community," "Wanted to be own boss," and "Always wanted to start own business."

Column 6 of Table 6a examines the same specification, restricting the sample to the four sectors with the highest usage rates of AI-related technologies. We find similar patterns, except that holding an advanced degree or previously owning a business do not have statistically significant coefficients in these select sectors. As to founding motivations (Column 2 of Table 6b), we find similar patterns and point estimates as for the broader startup sample, except that "Balance work and family" and "Best avenue for ideas/goods/services" are not statistically significant in explaining early growth among these more AI-intensive industries

4.3 Conditional Correlates of AI Adoption

We next look at how well markers of early growth potential predict AI use in 2017. Column 1 of Table 7 indicates that higher growth in the first few years of a startup's life is positively and significantly associated with AI use.[22] A 10pp increase in the average growth rate in the first three years is associated, *ceteris paribus*, with a 0.134pp increase in the likelihood of using AI—equivalent to a 2.2% increase when measured at the mean usage rate of 6% for the startup sample.

We then add the firm characteristics identified earlier, beginning with those of the primary owner, in column 2 of Table 7. The association between AI use and early revenue growth remains significant, albeit slightly lower at 0.128pp. Having an owner with an advanced degree is associated with a 1.04pp increase in the likelihood of using AI (a 17% increase relative to the mean). The probability of using AI is also higher for businesses with owners who previously owned a business (1.20pp). In contrast, mid-career owners (ages 35–54) display a negative (but statistically insignificant) conditional likelihood of adopting AI, while the relationship is more negative and

statistically significant for ages 55 and up. Lifestyle motivations are statistically insignificant in this specification.[23]

Columns 3 and 4 repeat this exercise, but instead focusing on startup financing and location, respectively. Early growth continues to play a role, though it further falls in Column 3. The presence of venture capital at startup or acquisition has a particularly strong relationship to AI use, it being associated with a 10.7pp increase in the probability of AI use (a 179% increase relative to the mean usage rate). Higher startup capitalization is also associated with a higher probability of AI use. The relationship between business location and AI use explored in Column 4 suggests that startups located in large CBSAs have a 0.74pp higher probability of using AI (or 12% of the mean rate of AI use) compared to those located in a micropolitan area with fewer than 50,000 people (the omitted group). The coefficient of early growth decreases only slightly compared to that in Column 1 once controls for location are included.

In Column 5, we include all previous covariates as well as innovation and business strategy characteristics. Process innovation, patents, and identifying IP protection as very important are the innovation and business strategy characteristics that have the strongest association with AI use, with probability increases of 5.4pp, 3.5pp, and 6.1pp, respectively. Firms that reported product innovation or a growth-oriented innovation strategy also have statistically significant increases in the likelihood of being AI users. Overall, we find that the presence of venture capital funding (which still increases the probability of AI use by 8.6pp), reliance on IP, process innovation, and patent ownership continue to have large and statistically significant associations with AI adoption, even conditional on a wide range of observables. Together, these four startup characteristics are associated with an increase in the probability of AI use by almost 24pp, equivalent to 4 times the startup sample's average rate of AI use. A depiction of the magnitudes for each of the items can be found in Figure 7. Note also that in Column 5 the association of location with AI use becomes weaker compared to Column 4, with no significant coefficient estimates for any of the CBSA size classes included. It is likely that concentration of innovative startups with younger owners and venture capital backing in large urban areas is a driver of this weaker correlation.

When all observables are included together in Column 5, early growth still has a statistically significant relationship with AI adoption, yet the magnitude falls to less than half of that in Column 1. A 10pp increase in early revenue growth is now associated with a 0.058pp increase in the

probability of AI use (equivalent to only a 1% increase at the mean usage rate).

Two observations are in order. The first is that a significant amount of the correlation between early growth and AI use is attributable to firm characteristics *associated* with growth, not necessarily to growth *per se*. A key advantage of our study lies in this ability to directly account for factors that have been associated with high-growth entrepreneurship, partialling out their direct relationship to AI use to identify likely drivers of AI use. The second is that the remaining association with growth early in the life cycle is consistent with the possibility that faster-growing startups may have the need and/or resources to adopt and use AI. Taken together, these findings point to greater growth of AI in what have historically been the most dynamic drivers of the aggregate economy.

Overall, our most richly specified model (Column 5) explains 23% of the variation in AI use, an increase of about 10% over the initial specification in Column 1. While the firm characteristics included in Column 5 contribute significantly to the explanatory power of the model, a substantial portion of the variation in AI adoption still remains unexplained, pointing to a large amount of unobserved heterogeneity underlying AI diffusion in the U.S.

Similar to our focus on startups because of firm age's negative relationship with AI use that we note in Section 3.4, we also look specifically at the four leading sectors for AI use (shown in Figure 2b): manufacturing, information, health care, and professional/technical services. In Column 6 we show that our results are also largely robust to focusing only on these 2-digit sectors that have the highest AI usage rates.

Finally, we consider our entire owner and revenue sample (column 3, Table 1) rather than restricting to startups. With this sample that extends back to 1997, we can include not only the average revenue growth for the initial years of the firm, but also the average revenue growth rate from the 3 most recent years for which revenue information is available, 2016–2018. Similar to Columns 1–5 in Table 7, the coefficient on early revenue growth weakens and loses significance as we include all owner characteristics, startup conditions, and innovation strategy objectives; however, the coefficient on the later-stage revenue growth remains statistically significant. This later period of revenue growth likely falls into post-adoption period for some firms, suggesting that the use of AI itself could have contributed to the improved performance of the firms who adopted AI—though we make no claims here of a causal link. While we report the complete set of results for this sample

in the online appendix Table A4, we report the fullest specification in Column 7 of Table 7. Here we find a statistically significant relationship between later-stage revenue growth and AI use, while the coefficient on early revenue growth is not statistically significant because of its correlation to owner characteristics, startup conditions, and innovation/business strategies.

Our analysis has identified some key firm-level characteristics that are significantly positively associated with early revenue growth, which turn out to be also highly positively correlated with AI use. Before summarizing our key results, we place them into context by recalling the properties of our startup sample used in Tables 6a, 6b, and 7. Specifically, these regressions leverage the subsample of young firms that report at least one owner with a nonzero share of the firm and have revenue information available—we thus necessarily exclude any publicly owned firms. Our tables and figures pertaining to the samples that do not include ownership information are not subject to these data restrictions. With these caveats in mind, we next summarize the key takeaways from our analysis.

5 Summary of Key Findings and Discussion

5.1 Diffusion of AI is low yet concentrated in key segments of the economy

Overall, the view of early AI use among U.S. firms is one of low diffusion, on average, yet with high concentration among certain sectors and among a smaller number of relatively large firms. The importance of the latter may be seen in that exposure to AI by workers is much higher than usage rates among firms: the firm-weighted rate of AI adoption rises from 5.8% to 18.2%, once employment weighting is incorporated.

These adoption rates are much lower than other studies which lack the representation and coverage of our data set, particularly with respect to smaller and medium-sized firms. While we do not directly observe impediments to AI adoption, scale certainly appears to play a role. Indeed, these low averages risk obscuring the fact that most very large firms (more than 5,000 employees) report at least some AI use, as well as higher intensity of use. The many smaller and medium-sized businesses we observe may not possess the vast amounts of data to develop and/or leverage AI applications, or they make lack the level of production necessary to justify early investments in AI.

This pattern of low yet skewed early AI adoption has important implications for what types of

firms are "at risk" of adoption, moving forward. If AI's niche is only a small group of firms that are specialized in certain industries or have already attained sufficient scale, then policy makers and managers need to update expectations regarding the ultimate diffusion and impact of AI. The disparity in adoption rates across firm types also suggest an "AI divide" among the population of businesses in the U.S—a small set of firms with special characteristics that we uncovered above lead the diffusion curve, with a potentially large number of laggards yet to overcome the impediments to AI adoption or to benefit from AI applications. That said, our detailed exploration into AI's leading edge indicates that the early adopters are the types of firms—innovative, high growth potential—that matter for economic growth and dynamism. Thus, their rising participation in the economy over time may bring with it greater presence and impact of AI in the overall economy.

5.2 Technology Interdependencies Matter

Availability of digital information (data) and computing resources (in this case, in the form of cloud computing) appears central to AI adoption across a broad swath of firms. In fact, while cloud computing has proven important for younger firms in prior work (Jin and McElheran 2017), the importance of cloud computing across the broader age and size distribution of U.S. firms is apparent in our more-representative statistics. Our findings in this study provide at least suggestive evidence that technological interdependencies, particularly with potential "enabling" technologies, need to be managed for proper deployment of AI. To the extent that it takes time, resources, and even economies of scale to accumulate key "enabling" technologies such as data and computing resources, considerable frictions to AI use likely remain.

5.3 Characterizing AI adoption poses a significant measurement challenge

An often under-emphasized point that emerges from our detailed analysis of AI adoption among startups (Table 7) is that typical observable firm characteristics leave unexplained a large fraction of the variation in AI adoption across firms. Even with high-dimensional controls (state–industry controls at the 6-digit NAICS), along with controls for firm age and owner gender, we can only explain around 20% of variation in adoption. This low explanatory power indicates that there remains significant unobserved heterogeneity underlying AI adoption among U.S. firms. Including additional, harder-to-measure factors from ABS related to leadership, capitalization, and innovative

strategies contributes significantly to the explanatory power (an increase of about 10% from the baseline specification). Nevertheless, a large amount of variation is left unexplained, pointing to the importance of idiosyncratic firm-level unobserved factors, such as the specific product or service being offered by a firm or the types of processes and tasks used in production that demand AI technologies. This points to important areas for follow-on data collection and research.

5.4 Characteristics of firm leaders predict AI use

Our interest in firm dynamics led us to examine AI use in a very large sample of startup firms for which the ABS provides unprecedented visibility to details of the organizational context. This revealed that more educated, more experienced, and younger owners (typically founders) are more likely to preside over AI adoption at their firms. One possible reason is that education may be a proxy for the complexity of products and services offered by the firm, which may require a higher degree of education—and, at the same time, advanced technology like AI—to be produced. For instance, a management consulting firm that offers predictive analytics services (a high-end business service) would more likely demand a Ph.D. degree for owners and managers.

Serial entrepreneurship, which has been linked to firm performance in prior work, also predicts AI use, suggesting a novel mechanism to performance gains in the digital age. Likewise, our finding that firms with owners younger than 35 are more likely to adopt AI is consistent with research finding that younger entrepreneurs tend to be more sophisticated users of advanced technology and may be more open to adopting these technologies in their businesses. This is younger than the mean age of high-growth founders (Azoulay et al. 2020) or recipients of the returns to digitalization (Barth et al. 2023) found in prior work.

Our finding that the motivations of owners and founders are correlated with AI use is quite novel to the literature on technology diffusion and its implications. High-growth and innovative firms are not only led by individuals motivated by bringing new ideas to market but also by those who report prosocial values such as helping or becoming involved in the community. In turn, such motivations are highly correlated with AI use. Other motivations for founding or owning firms matter, but there is a clear distinction between markers of growth-oriented and so-called "lifestyle" entrepreneurship when it comes to AI use. This again suggests an outsized impact of AI compared to its low early average adoption, while pointing to potential guardrails regarding its

specific applications, in practice.

5.5 Firms associated with "high-growth" entrepreneurship adopt AI

Among startups, early indicators of high growth potential (which we validate empirically in the rich panel of administrative data) are positively correlated with AI use. While higher initial capital levels at startup are associated with higher likelihood of AI use, firms receiving VC funding are substantially more likely to adopt AI,. VC-funded startups tend to be high-growth businesses even before VC-funding, and many venture capitalists have the skills and expertise to assess the underlying promise of a startup. Hence, VC funding could be a signal of firm potential not captured by other observables in our analysis. However, we also do not rule out reverse causality, whereby VC influence promotes AI adoption. Either way, the characteristics of VC-funded firms appear highly complementary to AI use in ways that have been heretofore unobserved.

That said, our ability to directly condition on a large number of markers of high-growth potential points to a robust relationship between AI use and growth *per se*, among startups. A possible interpretation is that firms with a demonstrated upward trajectory, early on, may have a greater demand (as well as the requisite resources and organizational complements) for AI. Even so, our analysis of startups, as well as of firms in which we can assess revenue growth later in life, points to an important relationship between growth and AI adoption that is consistent with performance gains arising from AI use. We cannot establish causality in our data, but hope that this robust finding provides the foundation for follow-on efforts to understand causality in these patterns.

5.6 Firm strategy matters for AI adoption

Our results also indicate a strong connection between AI adoption and a firm's innovative activity and innovation strategy. Both product and process innovation are positively correlated with AI adoption, but the latter has a much higher correlation. For example, startups with process innovation have a 5.4pp higher likelihood of adopting AI—a magnitude that is nearly equivalent to the overall usage rate (6%) of AI for startups. Similarly, firms that possess patents and those that identify IP protection as very important are more likely to adopt. These findings are consistent with the literature emphasizing firm-level complementarities between technology use and innovation.

5.7 Geographic disparity in the adoption of AI is pronounced

We find substantial geographic disparity in the adoption and use of AI among young single-unit firms and among single unit firms, in general. In particular, the rate of AI use by single-unit startups tends to be higher in CBSAs located in the southern and western parts of the U.S., technology hubs, such as Silicon Valley and the Research Triangle, and large metro areas such as San Francisco, Boston, Seattle, San Antonio, Nashville, San Diego, and Louisville.[24] The geographic disparity in the adoption of AI, combined with the high concentration of AI in large, innovative businesses (key findings 5.1 and 5.6) together suggest an "AI divide" that may be further reinforced if new AI-using, high-growth startups continue to be attracted to locations with high AI activity.

6 Conclusion

As AI advances and becomes more integrated into the workplace, there is a growing debate about whether it will increase productivity, what the effects on the workforce will be, or even whether AI will spark a revolution in business processes and models. Central to these debates is obtaining reliable measures of AI use across the U.S. firm population, identifying relevant adopter characteristics, and assessing worker exposure to AI.

The 2018 ABS technology module addresses this data need along a number of dimensions, providing detailed information about the early diffusion of AI and related technologies across a nationally representative set of firms, including detailed information on industry (at the NAICS6 level), age, and revenue over the firm life cycle. It further provides unprecedented insights into the founder and organizational characteristics for the set of AI adopting firms, who share many of the attributes that characterize high-growth entrepreneurship.

Among the many patterns we document, the potential for an "AI divide" emerges between different segments of the firm population. One is characterized by potential and motivations to expand, innovate, and transform the economy; and the other dominated by more lifestyle-oriented entrepreneurship with little prospects of growth and innovation. One is located squarely in an technology "hub," while the other is lacking the potential spillovers such clusters increasingly promise (regardless of urbanization). The gap between these segments is likely growing in part due to the use of advanced technologies like AI. Quantifying this dividing role of AI is a key challenge for

future work.

Our study is not without limitations. While our "headline" adoption statistics are robust and highly representative of the entire U.S. economy, our more nuanced characterization of AI users leans on a particular subsample of young firms. This sample is nevertheless large and covers important dimensions of the firm size and industry distributions, but it is also subject to non-trivial ownership and age restrictions. In particular, we trade off unusual visibility to early-stage entrepreneurial activity—as well as changes across the life cycle—for insights that are less applicable to the more-studied publicly traded firms dominating digital technology research. To the extent that we are trying to shed light on AI's future trajectory, rather than its past, this tradeoff is well worth making. Our findings are also conditional on survival: any firms that adopted AI and failed before 2018 are not represented in our data. This limitation is common in related work and underscores the importance of collecting data on new technologies as early as possible (e.g., McElheran 2018).

The patterns we identify arguably raise more questions than they answer. In particular, identifying causal relationships is beyond the reach of this first cross-section of data. Future collections can build a panel of stable measures or use instruments that pin down exogenous variation in AI use for that important endeavor. That said, the patterns we uncover concerning the who, what, and where of early adoption of AI contribute to a more evidence-based response to the emerging AI-driven business revolution that has received a high ratio of speculation to evidence. Our analysis demonstrates potential ways to overcome the measurement challenge and identifies paths for future work.

Endnotes

1 For reference, the quinquennial Economic Census typically has a response rate of approximately 80% for single-unit firms (Dinlersoz and Klimek 2011). The Community Innovation Survey in Europe reported response rates of 25–35% in Germany, resulting in approximately 5.5 thousand firms (Czarnitzki et al. 2023). The Eurostat survey on AI had response rates of 5–19% across 27 EU countries, yielding over 9.6 thousand firms (Hoffreumon et al. 2023).

2 The AI-related technologies covered are: automated-guided vehicles, machine learning, machine vision, natural language processing, and voice recognition. See Table 2. To best capture the extensive margin of AI use across industries, we combine usage rates across AI-related technologies. That said, our findings largely represent variation in the use of "machine learning" in production.

3 Studies of AI use and impacts outside of the U.S. have been on the rise, including in Europe (Czarnitzki et al. 2023; Hoffreumon et al. 2023), China (Lu et al. 2023; Beraja et al. 2023), Canada (Alexopoulos and Cohen 2018) and in the context of international trade (Goldfarb and Trefler 2018; Brynjolfsson et al. 2019).

4 E.g., Catalini et al. (2019); Akcigit et al. (2022); Lerner and Nanda (2020); Botelho et al. (2021).

5 See Brynjolfsson and Milgrom (2013) for more on complementarity theory and a review of the literature.

6 Such details are captured via responses from "primary owners" to the ABS, which excludes publicly traded firms, estates, trusts, government and tribal entities, associations, membership clubs, cooperatives and foreign entities.

7 Specifically, we recalculate the sample weights by stratifying the firms in the 2017 LBD and our final sample of firms in the ABS on firm size, age, and industry. These strata are defined by the 19 two-digit NAICS sectors and the 12 firm size and 12 firm age groups used in the BDS. We weight according to these more standard firm characteristics because of the unique sampling frames of the ABS (see Section B.1 in our data appendix).

8 Distributions taken from Zolas et al. (2020) show that nearly 70 percent of ABS firms, and more than 3/4 once weighted, have fewer than 10 employees, while the age distribution is more uniform.

9 Other technologies queried include augmented reality, robotics, touchscreens, automated storage and retrieval systems, and radio frequency identification (RFID). While some of these technologies may also contain an AI-related component, they are less likely to rely heavily on AI compared to the ones we consider to be AI-intensive. For instance, the 2018 ABS definition of robotics (a technology that has increasingly relied on AI in recent years) does not mention autonomy, but rather only reprogrammability (i.e., "traditional" industrial robots).

10 The rise in usage rates with firm size could be partly a "mechanical" function of scale. If we conceptualize larger firms as random collections of smaller ones and extrapolate usage rates as a function of number of employees, we would expect even higher usage rates among larger firms.

11 E.g., this controls for the distinction between manufacturing hardwood versus softwood veneer (NAICS 321211 vs. 321212) and bicycles vs. armored vehicles and tanks (336991 vs. 336992); newspaper publishers vs. greeting card publishers (513110 vs. 514191); and casino hotels versus bed-and-breakfast inns (721120 vs. 721191).

12 Since the LBD only extends back to 1976, the age distribution in our samples is truncated from above at 42 years, with all firms of age 42+ being assigned to the highest age percentile group. As a result, coefficient for the highest age percentile should be interpreted with caution.

13 We see similar patterns of increasing overlap as firms get larger. Figure A1 plots the share of AI and robotics adopters across the different size categories. Among AI adopters, the share of firms also adopting robotics rises with size. However, somewhat interestingly, we find the opposite pattern among robotics adopters. The share of robotics users that also use AI declines from 65.6% of robotics adopters in the smallest size category to less than half (49.3%) of firms with 20 or more employees. These patterns suggest that any underlying complementarities between AI and other technologies may have complex interactions with firm characteristics such as scale.

14 For details on these owner and firm characteristics collected in the ABS, see Section B.2 of our data appendix.

15 CBSAs are geographic areas that consist of one or more counties (or equivalents) with an urban center of at least 10,000 people plus adjacent counties that are socioeconomically tied to the urban center by commuting.

16 Because a sample of single-unit firms differs somewhat from our nationally representative baseline sample, we modify our firm weights. We do not, however, directly incorporate geographic information when generating our weights. For comments on incorporating geography into tabulation weights, see Section B.3 in our data appendix.

17 Because we only require location information, this sample includes startups that do not appear in our core startup sample from column 4, Table 1. However, because of the granularity of CBSAs and to protect confidentiality, we only report information for the 30,000 startups in the largest CBSAs (i.e., those with a population of one million or more).

18 Because Durham–Chapel Hill does not meet our population requirement to be designated as a "large" CBSA, this piece of the Research Triangle is absent from Figure 5a and Table A2. However, Figure 6a shows that Durham–Chapel Hill has a relatively high usage rate of AI when considering all single-unit firms.

19 We perform a similar analysis for all single-unit firms across 350+ CBSA categories (of all sizes—not just the largest) for which at least 20 single-unit AI firms exist. These can be found in the appendix Figures 6a and 6b. The results are qualitatively similar but with some interesting patterns of dispersion among the small and mid-sized CBSAs.

20 While it is well-known that size and age are positively correlated, many firms do not grow much with age.

21 Most studies of technology adoption and the impact of impact technology on performance suffer from similar survival bias (see Jin and McElheran 2017 for a discussion and evidence).

22 Note that we do not claim a causal relationship. In fact, faster-growing startups may be more able to use and benefit from AI if the initial costs of adoption or the scale of training data are large.

23 As described in Section 3.6.1, we find (unconditionally) that the use of AI-related technologies is negatively correlated with "Flexible hours" and "Balance work and family" reasons, and it is positively correlated with "Best avenue for ideas/goods/services" and "Help and/or become more involved in community" (Table A3). For simplicity we thus retain our "lifestyle" reason indicator in Table 7.

24 As shown in Figure 6a, when considering all single-unit firms there are also pockets of high adoption in some smaller CBSAs such as Provo, UT and Eugene, OR—including Madison, WI and Akron, OH when weighting by employment (see Figure 6b).

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Tables

Table 1: Summary Statistics of Firm Size and Age by Sample

	(1) ABS–LBD Linked Sample	(2) Baseline Sample	(3) Owner & Revenue Sample	(4) Startup Sample	(5) Startup Sample in Spec. Ind.
Employment (Unweighted)	89.32	61.69	17.68	10.39	9.11
Employment (Weighted)	26.28	20.77	9.78	7.4	6.91
Age (Unweighted)	16.33	16.60	8.94	2.87	2.87
Age (Weighted)	15.61	15.94	8.66	2.82	2.83
Observations (Unweighted)	573,000	447,000	209,000	75,000	28,000
Observations (Weighted)	5,180,000	4,050,000	1,970,000	740,000	228,000

Notes: Tabulations based on 2018 ABS data linked with the 2017 Longitudinal Business Database (LBD) data for size and age. Firms that did not respond to any of the 2018 ABS survey are excluded. See Zolas et al. (2020) for further details and breakdown of the size and age distribution for the full sample of linked ABS–LBD firms. Weights are computed by stratifying the firms in the 2017 LBD and our final sample of firms in the ABS on firm size, age, and industry. These strata are defined by the 19 two-digit NAICS sectors and the 12 firm size and 12 firm age groups used in the BDS. Samples in columns 2–4 are restricted by needing to contain information on the primary owner and are born on or after 1997, the first year that the LBD contains firm-level revenue measures. "Startups" are defined as being born on or after 2012. Specific Industries refers to a subset of 2-digit NAICS sectors that are more engaged in AI usage: Manufacturing ("31–33"), Information ("51"), Professional and Scientific Services ("54") and Healthcare ("62").

Table 2: Definitions of Technologies

Augmented reality	Technology that provides a view of a real-world environment with computer-generated overlays.
Automated guided vehicles (AGV) or AGV systems	A computer-controlled transport vehicle that operates without a human driver. AGVs navigate facilities through the use of software and sensors.
Automated storage and retrieval systems	Technology that locates, retrieves, and replaces items from predetermined storage locations.
Machine learning	Computer algorithms that use data to improve their predictive performance without being reprogrammed.
Machine vision	Technology used to provide image-based automatic inspection, recognition or analysis.
Natural language processing	Technology that allows a computer to process human speech or text.
Radio-frequency identification (RFID) system	A system of tags and readers used for identification and tracking. Tags store information and transmit them using radio waves. Readers maybe be mobile or fixed in place.
Robotics	Reprogrammable machines capable of automatically carrying out a complex set of actions.
Touchscreens/kiosks for customer interface (Examples: self-checkout, self-check-in, touchscreen ordering)	A computer with a touchscreen that allows a customer to receive information or perform tasks related to the business such as registering for a service or purchasing items.
Voice recognition software	Software that converts speech to text or executes simple commands based on a limited vocabulary or executes more complex commands when combined with natural language processing.

Notes: "AI-related" technologies for the purposes of this study are in bold. Definitions validated during cognitive testing of the survey instrument. See Zolas et al. (2020) for further details.

Table 3: Top Non-manufacturing Industries (4-digit NAICS) for AI Use

NAICS	NAICS Description	Mean (All Industries)	0.058
6215	Medical and Diagnostic Laboratories		0.237
6211	Offices of Physicians		0.190
5112	Software Publishers		0.156
5415	Computer Systems Design and Related Services		0.141
5182	Data Processing, Hosting, and Related Services		0.139
5191	Other Information Services		0.126
6214	Outpatient Care Centers		0.121
5411	Legal Services		0.120
1151	Support Activities for Crop Production		0.119
4248	Beer, Wine, and Distilled Alcoholic Beverage Merchant Wholesalers		0.112

Notes: Tabulated from the ABS–LBD linked sample (column 1, Table 1). Shares computed using the LBD tabulation weights of firm counts, divided by the total number of firms (including those that responded with "Don't Know" or missing), then scaled up by the total number of non-missing and "Don't Know" responses for each technology. The 2017 industry figures from the LBD are the figures listed in the tables. Industry tabulations for multi-unit firms are generated from the largest payroll industry within the firm (if there is a tie, then the industry with the most employment is used).

Table 4: Correlates of AI Use - Firm Size, Age, and Other Technology Use

	(1)	(2)
Description	**Use AI**	**Use AI**
Emp. pctile 51–75	0.011***	0.004***
	(0.001)	(0.001)
Emp. pctile 76–90	0.025***	0.012***
	(0.001)	(0.001)
Emp. pctile 91–95	0.036***	0.019***
	(0.002)	(0.002)
Emp. pctile 96–99	0.053***	0.032***
	(0.002)	(0.002)
Emp. pctile 99+	0.086***	0.063***
	(0.004)	(0.004)
Age pctile 51–75	-0.006***	-0.001
	(0.001)	(0.001)
Age pctile 76–90	-0.011***	-0.004***
	(0.001)	(0.001)
Age pctile 91–95	-0.015***	-0.006***
	(0.002)	(0.002)
Age pctile 96–99	-0.022***	-0.013***
	(0.003)	(0.003)
Age pctile 99+	-0.008***	-0.001
	(0.002)	(0.002)
Use: Digitization		0.019***
		(0.001)
Use: Cloud		0.050***
		(0.001)
State-by-Ind Controls	Yes	Yes
Observations (rounded)	447,000	447,000
R-squared	0.102	0.115

Notes: Standard errors in parentheses. *, ** and *** denotes significance at the 10, 5 and 1% respectively. The coefficient estimates represent the relative AI usage rates across size and age percentiles within a 6-digit (NAICS) industry, by state (State-by-Ind controls). The underlying sample is the baseline sample (column 2, Table 1).

Table 5: Firm Characteristics by AI Use Status (%)
Startup Sample

	Full Sample	Use AI	Do not use AI	Difference
Primary Owner Characteristics				
Hold Advanced Degree	22.4	31.5	21.9	9.6
Owned Prior Business	41.4	43.8	41.2	2.6
Missing Age	7.4	9.4	7.2	2.2
Owner Age (0–34)	11.5	11.2	11.5	-0.3
Owner Age (35–54)	54.5	55.1	54.5	0.6
Owner Age (55+)	26.6	24.7	26.8	-2.1
"Very Important" Reasons for Owning the Business				
Wanted to be own boss	63.9	62.9	64	-1.1
Flexible Hours	54.5	52.8	54.7	-1.9
Balance work and family	56.4	55.1	56.4	-1.3
Opportunity for greater income	63.0	62.9	63	-0.1
Best avenue for ideas/goods/services	56.9	61.8	56.7	5.1
Unable to find employment	7.2	9.0	7.1	1.9
Working for someone else not appealing	30.5	31.5	30.5	1.0
Always wanted to start own business	48.5	49.4	48.5	0.9
Entrepreneurial role model	24.5	27.0	24.3	2.7
Carry on family business	9.9	11.5	9.9	1.6
Help or become involved in community	24.1	30.3	23.7	6.6
Other reasons	8.0	10.6	7.8	2.8
Startup Financing				
Funded by VC	0.9	2.9	0.8	2.1
Missing Startup Capitalization	8.4	6.3	8.5	-2.2
Startup Capitalization <25K	38.4	38.2	38.4	-0.2
Startup Capitalization 25K–1M	38.9	42.7	38.8	3.9
Startup Capitalization 1M+	2.5	3.8	2.4	1.4
Don't Know Startup Capitalization	11.8	10.3	11.9	-1.6
Innovation & Business Strategy				
Process Innovation	19.9	39.3	18.7	20.6
Product Innovation	53.6	66.3	52.8	13.5
Patents Owned or Pending	2.1	5.2	1.9	3.3
IP is very important	19.5	40.4	18.1	22.3
Growth-oriented innovation strategy	69.5	77.5	68.9	8.6
Geography				
In Micropolitan or Rural CBSA ($< 50,000$)	22.8	20.0	23.0	-3.0
In Small-Sized CBSA ($< 250,000$)	9.1	8.1	9.1	-1.0
In Medium-Sized CBSA ($< 1,000,000$)	17.8	17.1	17.8	-0.7
In Large-Sized CBSA ($1,000,000+$)	50.3	55.1	49.9	5.2

Notes: "Use" is defined as having responded with "In use for less than 5% of production or service", "In use for between 5%–25% of production or service" or "In use for more than 25% of production or service" for the category listed on any of the AI-based Business Technologies. The "Growth-oriented innovation strategy" indicator is equal to 1 if a firm responded that a focus on introducing new goods or services, reaching new customer groups, or opening up new domestic or export markets was a "very important" innovation strategy for the business. The underlying sample is our startup sample (column 4, Table 1).

Table 6a: Correlates of Early Firm Growth

Description	(1) Revenue Growth (First 3 years)	(2) Revenue Growth (First 3 years)	(3) Revenue Growth (First 3 years)	(4) Revenue Growth (First 3 years)	(5) Revenue Growth (First 3 years)	(6) Specific Industries Revenue Growth (First 3 years)
Advanced Degree (1/0)	0.0248***			0.0161**	0.0167**	0.0129
	(0.0069)			(0.0068)	(0.0068)	(0.0108)
Prior Business (1/0)	0.0194***			0.0094**	0.0076*	0.0107
	(0.0045)			(0.0045)	(0.0045)	(0.0079)
Owner Age (35-54)	-0.0463***			-0.0446***	-0.0428***	-0.0445***
	(0.0070)			(0.0070)	(0.0070)	(0.0134)
Owner Age (55+)	-0.1009***			-0.0961***	-0.0937***	-0.1037***
	(0.0079)			(0.0079)	(0.0079)	(0.0151)
Lifestyle Reason (1/0)	-0.0037			-0.0081*		
	(0.0049)			(0.0049)		
Funded by Venture Capital (1/0)		0.1607***		0.1289***	0.1276***	0.1897***
		(0.0364)		(0.0362)	(0.0362)	(0.0633)
Startup Capitalization 25K-1M (1/0)		0.0629***		0.0564***	0.0553***	0.0965***
		(0.0051)		(0.0051)	(0.0051)	(0.0095)
Startup Capitalization 1M+ (1/0)		0.1393***		0.1308***	0.1295***	0.2109***
		(0.0173)		(0.0172)	(0.0172)	(0.0370)
Process Innovation (1/0)			0.0397***	0.0352***	0.0328***	0.0290***
			(0.0056)	(0.0056)	(0.0056)	(0.0096)
Product Innovation (1/0)			0.0132***	0.0100**	0.0093**	0.0157**
			(0.0043)	(0.0043)	(0.0043)	(0.0078)
Patents Owned or Pending (1/0)			0.0773***	0.0591***	0.0586***	0.0621**
			(0.0207)	(0.0205)	(0.0204)	(0.0312)
IP is very important (1/0)			0.0402***	0.0327***	0.0302***	0.0327***
			(0.0061)	(0.0061)	(0.0061)	(0.0105)
Growth-oriented innovation strategy (1/0)			0.0542***	0.0480***	0.0402***	0.0368***
			(0.0047)	(0.0047)	(0.0048)	(0.0084)
State-by-Ind Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm Age Controls	Yes	Yes	Yes	Yes	Yes	Yes
Owner Gender Controls	Yes	Yes	Yes	Yes	Yes	Yes
Reasons for Owning Controls	No	No	No	No	Yes	Yes
Observations (rounded)	75,000	75,000	75,000	75,000	75,000	28,000
R-squared	0.227	0.229	0.230	0.237	0.238	0.228

Notes: Standard errors in parentheses. *, ** and *** denotes significance at the 10, 5 and 1% respectively. Revenue growth refers to the three-year average of the log-difference measure of annual revenue growth. State-by-Ind controls refers to state and 6-digit NAICS industry dummies. The underlying sample for Columns 1–5 is our startup sample (column 4, Table 1). Specific Industries in Column 6 refers to the startup sample in specific industries (column 5, Table 1). For a detailed breakdown of "Reasons for Owning Controls" (which replaces "Lifestyle Reason" in Columns 5 and 6), please refer to Table 6b.

Table 6b: Correlates of Early Firm Growth
(Detailed Reasons for Owning Business)
Corresponds to Columns 5 and 6 in Table 6a

Description	(1) Revenue Growth (First 3 years)	(2) Spec. Ind. Revenue Growth (First 3 years)
Wanted to be own boss (1/0)	0.0199***	0.0291***
	(0.0062)	(0.0108)
Flexible hours (1/0)	-0.0175***	-0.0303***
	(0.0061)	(0.0108)
Balance work and family (1/0)	-0.0115*	-0.0103
	(0.0060)	(0.0106)
Opportunity for greater income (1/0)	-0.0002	0.0015
	(0.0058)	(0.0094)
Best avenue for ideas/goods/services (1/0)	0.0139***	0.0127
	(0.0052)	(0.0091)
Unable to find employment (1/0)	-0.0252***	-0.0317**
	(0.0081)	(0.0158)
Working for someone else not appealing (1/0)	-0.0006	-0.0052
	(0.0051)	(0.0092)
Always wanted to start own business (1/0)	0.0105*	0.0173*
	(0.0054)	(0.0095)
Entrepreneurial role model (1/0)	-0.0009	-0.0138
	(0.0057)	(0.0107)
Carry on family business (1/0)	-0.0029	0.0086
	(0.0076)	(0.0177)
Help or become involved in community (1/0)	0.0212***	0.0223**
	(0.0055)	(0.0096)
Other reasons (1/0)	0.0141*	0.0108
	(0.0079)	(0.0145)

Notes: This table breaks out the individual "Reasons for Owning Controls" and directly corresponds to Table 6a. Specifically, Column 1 belongs with Column 5 in Table 6a and Column 2 belongs with Column 6 in the same table. Standard errors in parentheses. *, ** and *** denotes significance at the 10, 5 and 1% respectively. Revenue growth refers to the three-year average of the log-difference measure of annual revenue growth. State-by-Ind controls refers to state and 6-digit NAICS industry dummies. The underlying sample for Columns 1–5 is our startup sample (column 4, Table 1). Specific Industries in Column 6 refers to the startup sample in specific industries (column 5, Table 1).

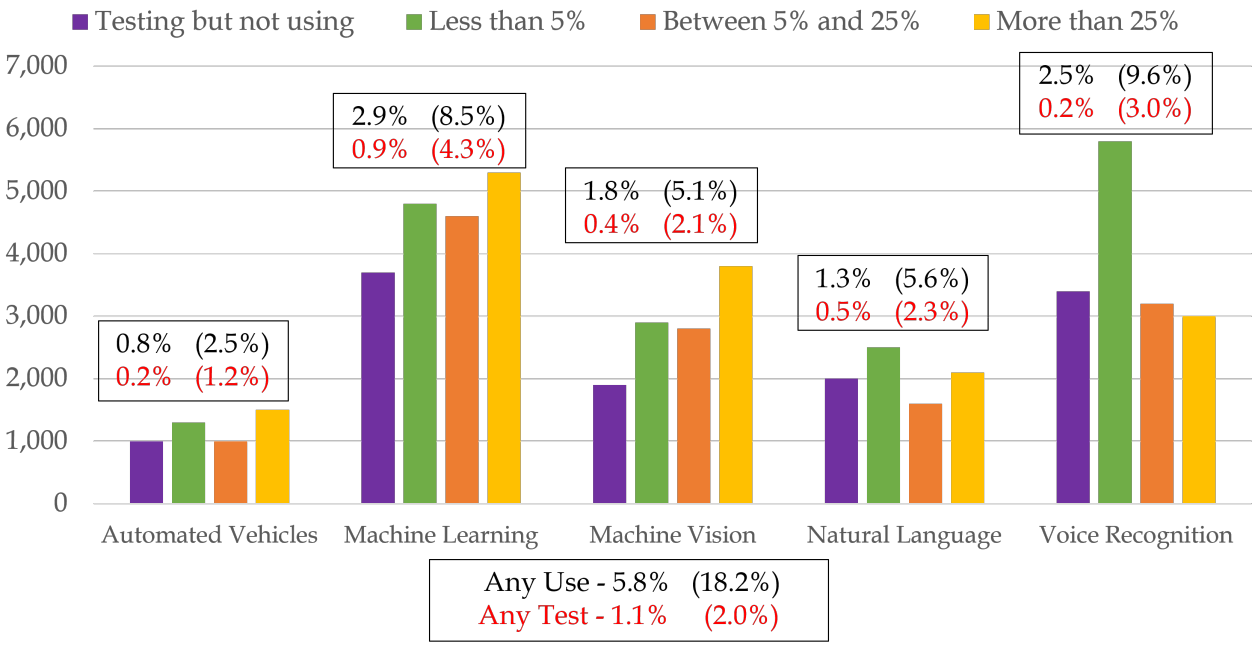
Table 7: Correlates of AI Adoption – including Early Firm Growth Rate

	(1)	(2)	(3)	(4)	(5)	(6) Specific Industries	(7) Full Owner & Rev. Sample
Description	**Use AI**	**Use AI**	**Use AI**	**Use AI**	**Use AI**	**Use AI**	**Use AI**
Revenue Growth (First 3 years)	0.0134***	0.0128***	0.0112***	0.0133***	0.0058**	0.0150***	0.0021
	(0.0024)	(0.0024)	(0.0024)	(0.0024)	(0.0024)	(0.0045)	(0.0016)
Revenue Growth (Last 3 years)							0.0041**
							(0.0018)
Advanced Degree (1/0)		0.0104***			0.0043	0.0159***	0.0087***
		(0.0032)			(0.0031)	(0.0054)	(0.0018)
Prior Business (1/0)		0.0120***			0.0066***	0.0081*	0.0050***
		(0.0023)			(0.0023)	(0.0047)	(0.0013)
Owner Age (35-54)		-0.0029			0.0016	0.0046	0.0004
		(0.0034)			(0.0033)	(0.0069)	(0.0026)
Owner Age (55+)		-0.0092**			-0.0019	0.0008	-0.0014
		(0.0038)			(0.0038)	(0.0078)	(0.0027)
Lifestyle Reason (1/0)		-0.0004			-0.0032	-0.0158***	-0.0021
		(0.0024)			(0.0024)	(0.0052)	(0.0013)
Funded by Venture Capital (1/0)			0.1071***		0.0862***	0.1048***	0.0578***
			(0.0161)		(0.0156)	(0.0254)	(0.0101)
Startup Capitalization 25K-1M (1/0)			0.0148***		0.0078***	0.0121**	0.0090***
			(0.0028)		(0.0027)	(0.0058)	(0.0015)
Startup Capitalization 1M+ (1/0)			0.0297***		0.0139*	0.0129	0.0130***
			(0.0078)		(0.0078)	(0.0159)	(0.0044)
Process Innovation (1/0)					0.0544***	0.0614***	0.0574***
					(0.0034)	(0.0062)	(0.0020)
Product Innovation (1/0)					0.0078***	0.0197***	0.0090***
					(0.0022)	(0.0045)	(0.0012)
Patents Owned or Pending (1/0)					0.0353***	0.0361**	0.0329***
					(0.0115)	(0.0160)	(0.0064)
IP is very important (1/0)					0.0605***	0.0656***	0.0572***
					(0.0037)	(0.0063)	(0.0021)
Growth-oriented innovation strategy (1/0)					0.0066***	0.0058	0.0091***
					(0.0023)	(0.0047)	(0.0012)
Small CBSA				-0.0006	-0.0007	-0.0010	0.0003
				(0.0043)	(0.0043)	(0.0101)	(0.0022)
Medium CBSA				0.0003	-0.0010	0.0087	-0.0004
				(0.0036)	(0.0036)	(0.0082)	(0.0019)
Large CBSA				0.0074**	0.0042	0.0075	0.0013
				(0.0031)	(0.0031)	(0.0070)	(0.0017)
State-by-Ind Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Age Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Owner Gender Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations (rounded)	75,000	75,000	75,000	75,000	75,000	28,000	209,000
R-squared	0.209	0.210	0.211	0.209	0.230	0.228	0.148

Notes: Standard errors in parentheses. *, ** and *** denotes significance at the 10, 5 and 1% respectively. Revenue growth refers to the three-year average of the log-difference measure of annual revenue growth. "Use" is defined as having responded with "In use for less than 5% of production or service", "In use for between 5%–25% of production or service" or "In use for more than 25% of production or service" for the category listed on any of the AI-based Business Technologies (Automated Guided Vehicles, Natural Language Processing, Machine Learning, Machine Vision and Voice Recognition). State-by-Ind controls refers to state-specific 6-digit NAICS industry dummies. The underlying sample for Columns 1–5 is our startup sample (column 4, Table 1). Specific Industries in Column 6 refers to the startup sample in specific industries (column 5, Table 1). Full Owner & Rev. Sample in Column 7 refers to our owner and revenue sample (column 3, Table 1). Column 7 is identical to column 5 of Table A4, which contains columns similar to this table for the full owner and revenue sample.

Figures

Figure 1: Prevalence of AI-Related Business Technologies (AI) in the United States by 2017



Note: Tabulations (bars) based on unweighted and non-imputed responses from the full ABS–LBD linked sample. "Don't Know" and missing responses are excluded from this figure. The figures listed above the bars reflect the firm-weighted usage rates, followed by employment-weighted usage rates in parentheses (both non-imputed and from our baseline sample). The figures in red font indicate the "Testing" rate for each technology. "Use" is defined as having responded with "In use for less than 5% of production or service", "In use for between 5%–25% of production or service" or "In use for more than 25% of production or service" in the "Business Technologies" module of the ABS. Shares are computed using the LBD tabulation weights of firm counts divided by the total number of firms (excluding those that responded with "Don't Know" or missing). Employment weights are calculated by multiplying the LBD tabulation weights by firm employment.

AI Use Intensity and Testing Rates

Figure 2a: By Firm Size

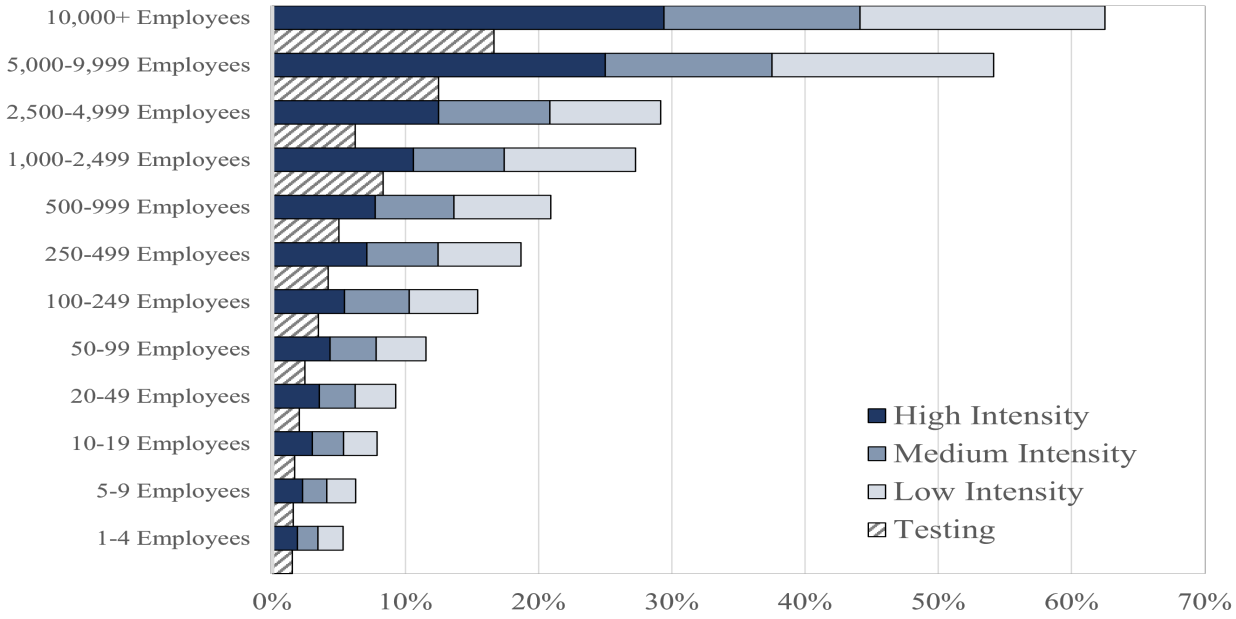
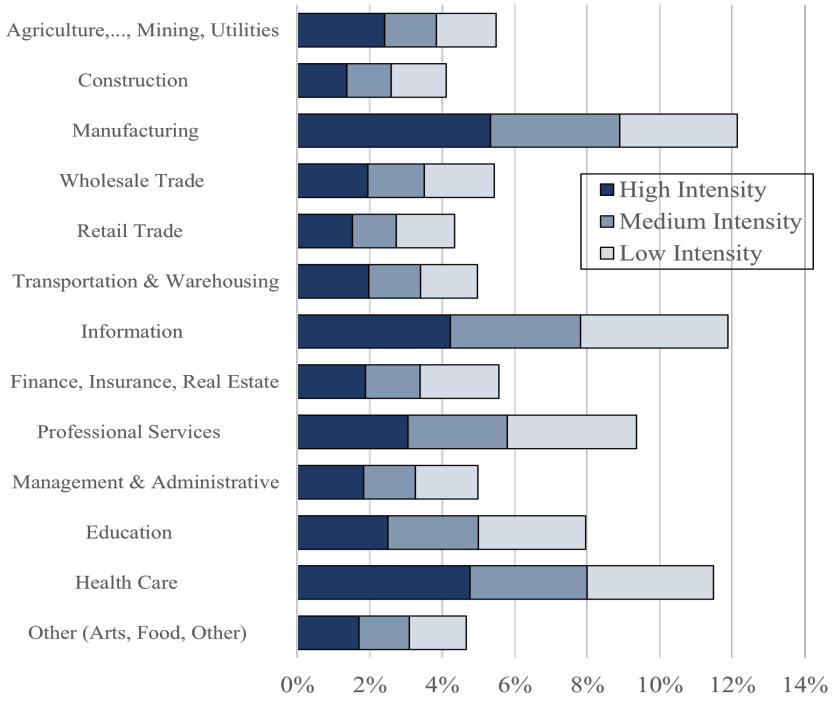
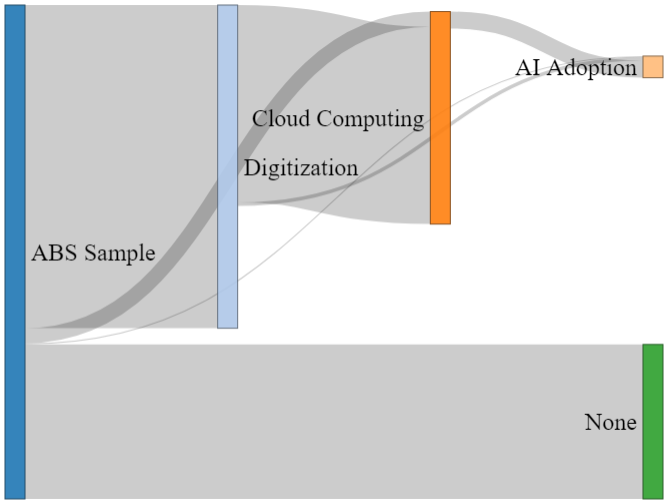


Figure 2b: By Sector



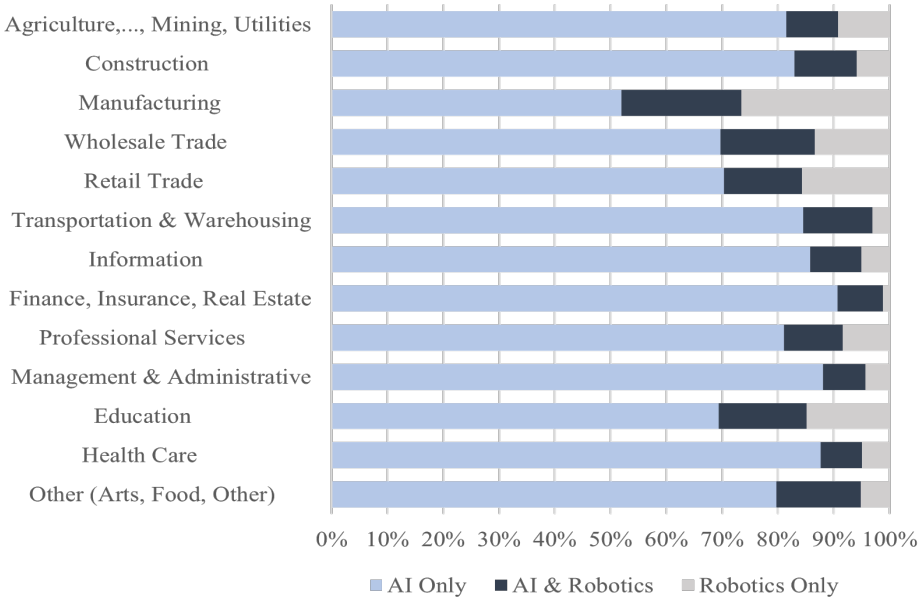
Notes: These figures visually represent the weighted share of firms that indicate intensity of use of at least one of the following business technologies: Automated Guided Vehicles, Machine Learning, Machine Vision, Natural Language Processing, or Voice Recognition (see Table 2) Values are based on the imputed probabilities for respondents who answered "Missing" or "Don't Know" to one or more of the aforementioned business technologies. High intensity corresponds to respondents utilizing at least one of the AI-based business technologies "In use for more than 25% of production or service." Medium intensity corresponds to "In use for between 5%–25% of production or service." Low intensity corresponds to "In use for less than 5% of production or service."

Figure 3: Technology Interdependencies



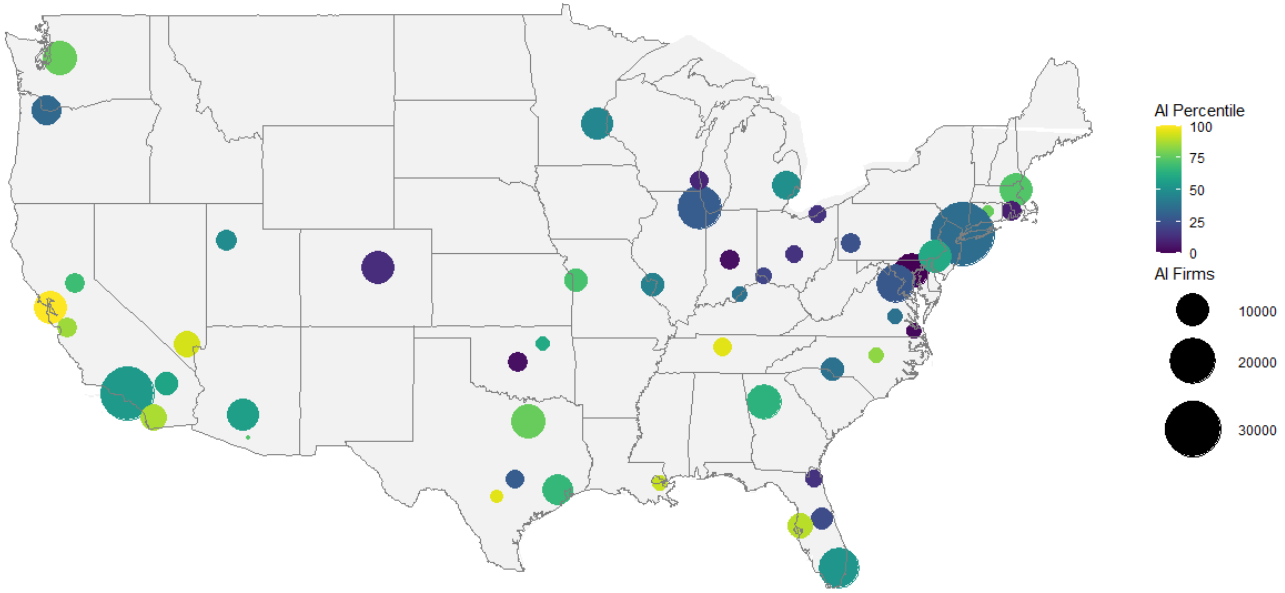
Notes: Sankey representation of firm counts in the ABS–LBD linked sample (column 1, Table 1) as they progress from no technology adoption to reliance on digital information, then cloud computing, and finally AI adoption. The size of the grey area is representative of the number of firm progressing to the next "stage" of technology use. Note that the calculations are made using imputed values for missing responses. Use of AI constitutes the weighted count of firms that indicate use of at least one of the following business technologies: Automated Guided Vehicles, Machine Learning, Machine Vision, Natural Language Processing and Voice Recognition (Table 2).

Figure 4: AI and Robotics Use by Sector



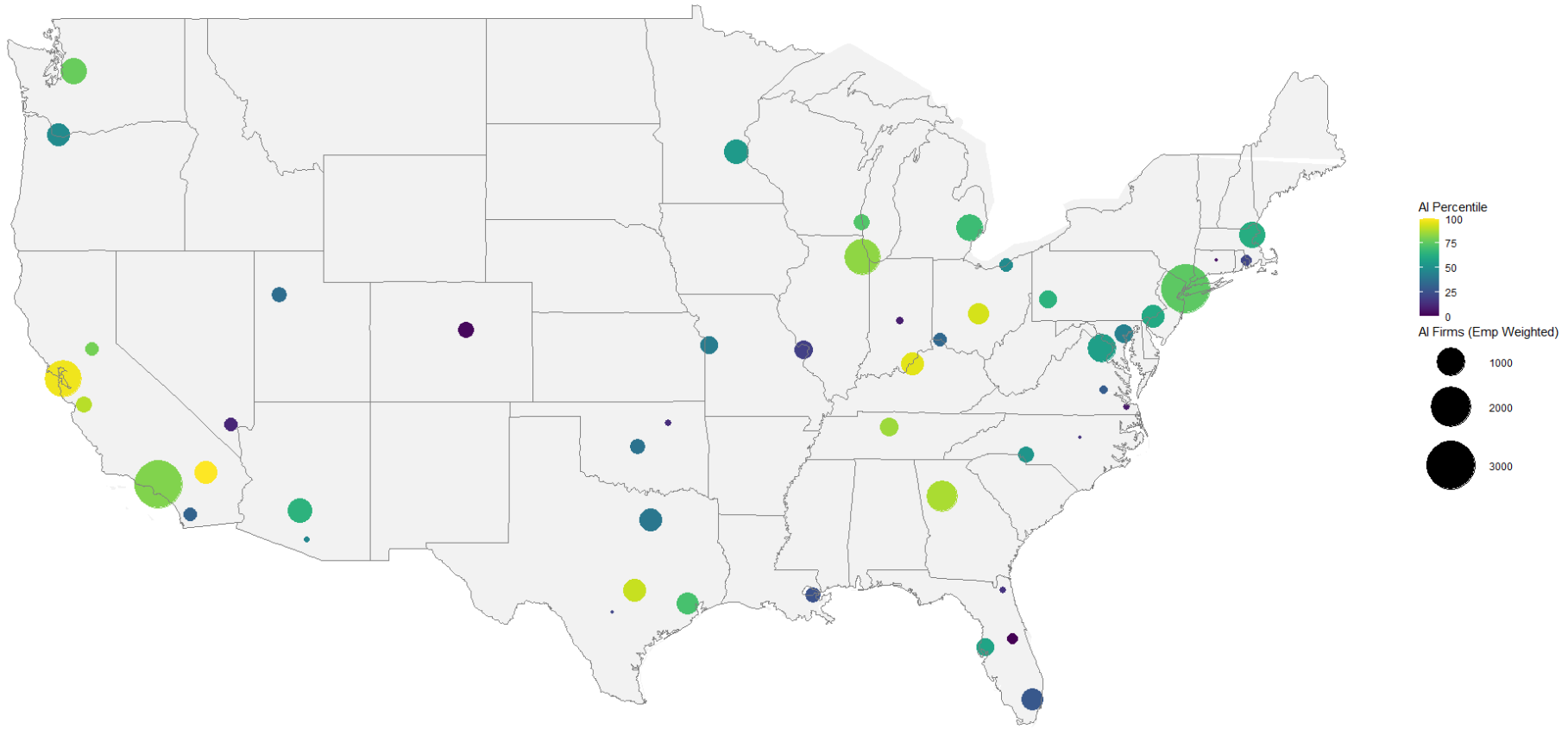
Notes: Percentage of firms that use AI or robotics among firms in the ABS–LBD linked sample that also reported using either an AI-based technology or robotics (roughly 34,000 firms). AI use is defined by the weighted share of firms by sector that indicate use of at least one of the following business technologies: Automated Guided Vehicles, Machine Learning, Machine Vision, Natural Language Processing and Voice Recognition. Robotics use is collected from the same survey question (see Table 2) and similarly weighted.

Figure 5a: AI Use Rates by Single-Unit Startups across Large Core-Based Statistical Areas (CBSAs)



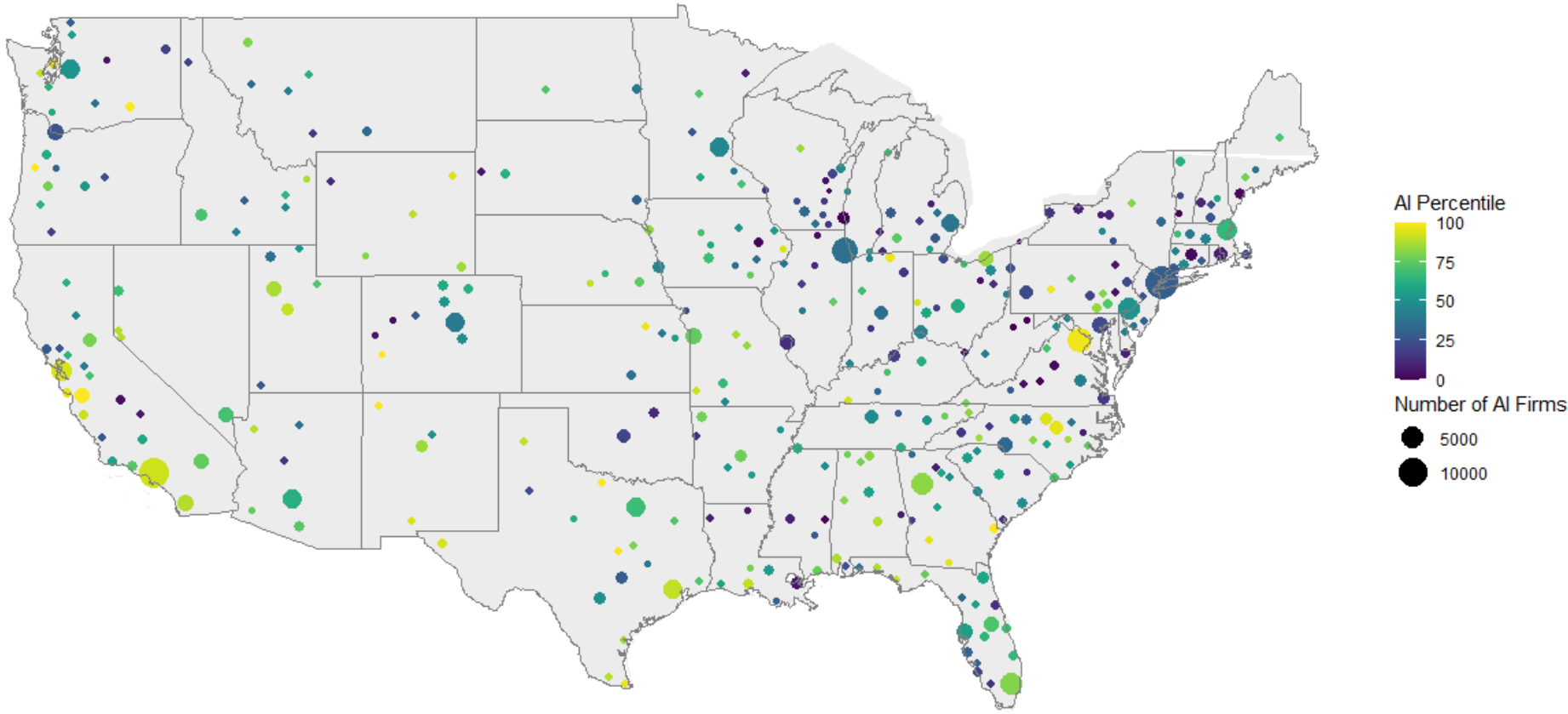
Notes: The map includes the largest (population $> 1,000,000$) Core-Based Statistical Areas (CBSAs). Bubble sizes in the map represent the (firm-weighted) number of single-unit startups (i.e., firm age ≤ 5 years) that use AI within each CBSA, and the color gradient represents the percentile rank of the usage rate of AI across single-unit startups within the largest CBSAs. Unlike our "startup sample" described in Table 1 (and used for regressions in Tables 6a, 6b, and 7), the underlying sample of this figure is not restricted to firms with owner or revenue information. Rather, the underlying sample is a subset of our baseline sample—specifically, single-unit startups located in large CBSAs (roughly 30,000 firms).

Figure 5b: AI Use Rates by Young Single-Unit Firms across Large Core-Based Statistical Areas (CBSAs) - Employment-weighted



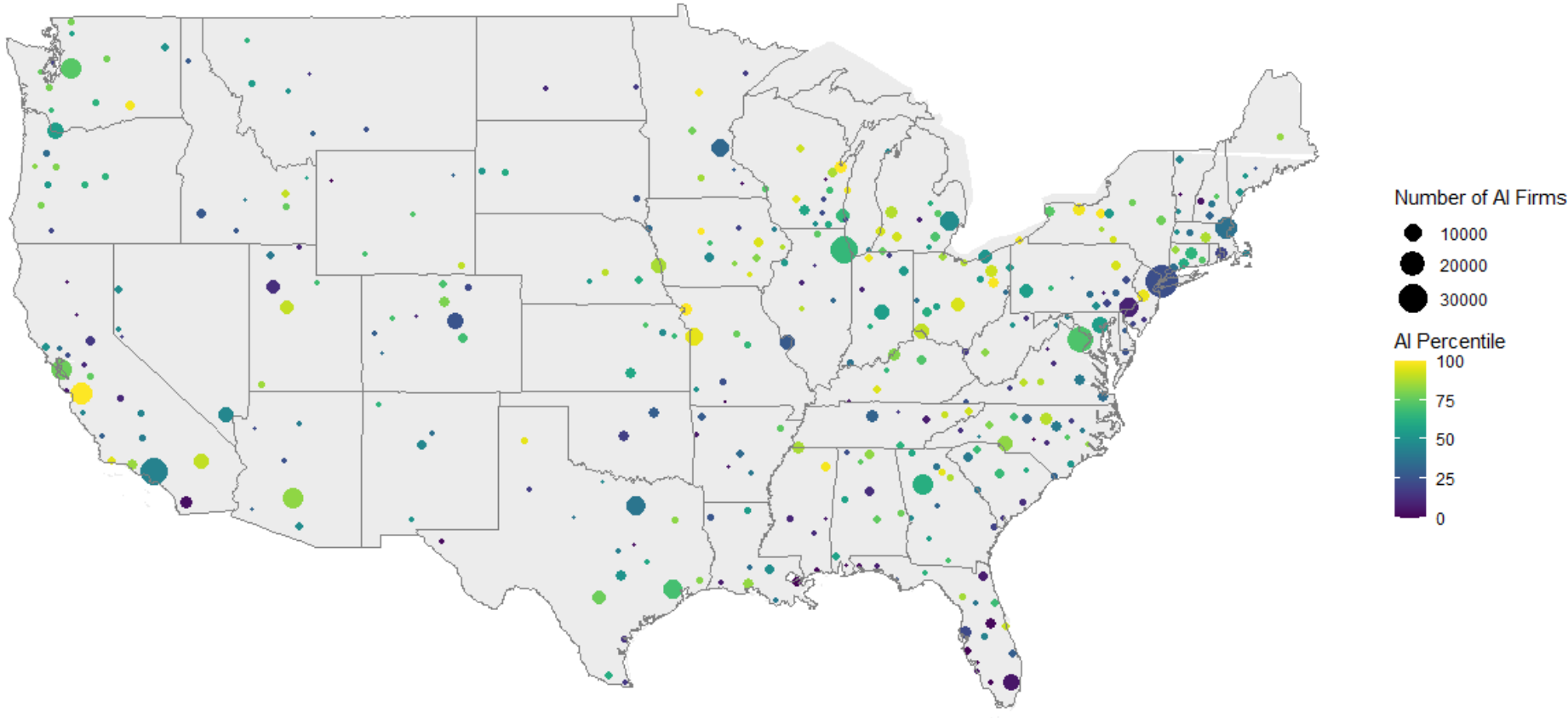
Notes: The map includes the largest (population $> 1,000,000$) Core-Based Statistical Areas (CBSAs). Bubble sizes in the map represent the employment-weighted number of single-unit startups (i.e., firm age ≤ 5 years) that use AI within each CBSA, and the color gradient represents the percentile rank of the usage rate of AI across single-unit startups within the largest CBSAs. Unlike our "startup sample" described in Table 1 (and used for regressions in Tables 6a, 6b, and 7), the underlying sample of this figure is not restricted to firms with owner or revenue information. Rather, the underlying sample is a subset of our baseline sample—specifically, single-unit startups located in large CBSAs (roughly 30,000 firms).

Figure 6a: AI Use Rates by Single-Unit Firms across Core-Based Statistical Areas (CBSAs)



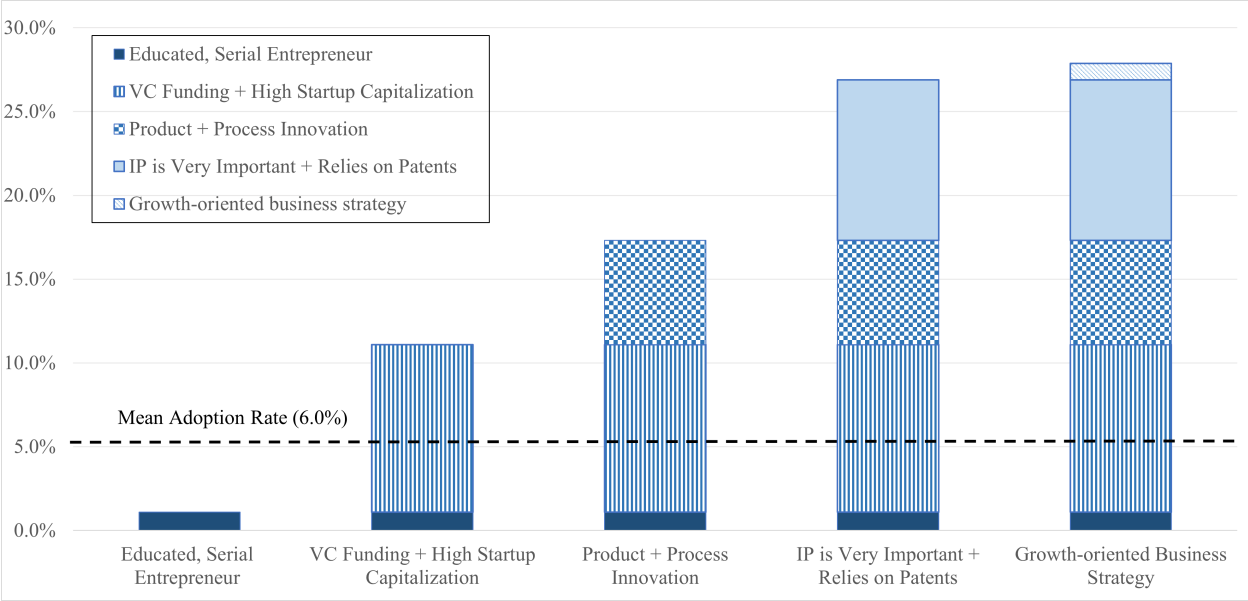
Notes: The map includes the 350 Core-Based Statistical Areas (CBSAs) with at least 20 single-unit firms using AI. Bubble sizes in the map represent the (firm-weighted) number of single-unit firms from our baseline sample that use AI within each CBSA, and the color gradient represents the percentile rank of the adoption rate of AI across single-unit firms within each CBSA type: Micropolitan ($< 50,000$ persons), Small (50,000–250,000 persons), Medium Small (250,000–1,000,000 persons), and Large ($> 1,000,000$ persons).

Figure 6b: AI Use Rates by Single-Unit Firms across Core-Based Statistical Areas (CBSAs) - Employment-weighted



Notes: The map includes the 350 Core-Based Statistical Areas (CBSAs) with at least 20 single-unit firms using AI. Bubble sizes in the map represent the employment-weighted number of single-unit firms from our baseline sample that use AI within each CBSA, and the color gradient represents the percentile rank of the adoption rate of AI across single-unit firms within each CBSA type: Micropolitan ($< 50,000$ persons), Small (50,000–250,000 persons), Medium Small (250,000–1,000,000 persons), and Large ($> 1,000,000$ persons).

Figure 7: Probability of AI Use : Select Marginal Effects (Linear Probability Model)



Notes: This figure plots select coefficient values from Column 5 in Table 7 to highlight the cumulative effects of key covariates in determining whether or not a firm adopts AI according to the definition of "use" in Table 7. The variables were grouped as follows: For "Educated, Serial Entrepreneurs", we combined the coefficients for "Advanced Degree" and "Prior Business." For "VC Funding + High Startup Capitalization", we combined the coefficients for "Funded by Venture Capital" and "Startup Capitalization 1M+". For "Product + Process Innovation", we combined the coefficients for "Process Innovation" and "Product Innovation". For "IP is Very Important + Relies on Patents", we combined the coefficients for "IP is very important and "Patents Owned or Pending". Finally, for "Growth-oriented Business Strategy", we combined the coefficient for "Growth-oriented innovation strategy" and took the negative of "Lifestyle Reason". The mean adoption rate is the mean AI usage rate for our startup sample (column 4, Table 1).

A Appendix Tables and Figures

Appendix Tables

Table A1: Firm Characteristics non-manufacturing industries for specific AI Technologies

Automated Guided Vehicles (AGV)			
NAICS	NAICS Description	Mean (All Industries)	0.008
1151	Support Activities for Crop Production		0.074
4245	Farm Product Raw Material Merchant Wholesalers		0.048
2379	Other Heavy and Civil Engineering Construction		0.037
Machine Learning			
NAICS	NAICS Description	Mean (All Industries)	0.029
5112	Software Publishers		0.103
5182	Data Processing, Hosting, and Related Services		0.084
5415	Computer Systems Design and Related Services		0.082
Machine Vision			
NAICS	NAICS Description	Mean (All Industries)	0.018
6115	Technical and Trade Schools		0.048
5112	Software Publishers		0.041
5415	Computer Systems Design and Related Services		0.040
Natural Language Processing			
NAICS	NAICS Description	Mean (All Industries)	0.013
5112	Software Publishers		0.063
5191	Other Information Services		0.059
5415	Computer Systems Design and Related Services		0.056
Voice Recognition			
NAICS	NAICS Description	Mean (All Industries)	0.026
6215	Medical and Diagnostic Laboratories		0.199
6211	Offices of Physicians		0.159
5411	Legal Services		0.093

Notes: Tabulated from the ABS–LBD linked sample (column 1, Table 1).). Shares are computed using the LBD tabulation weights of firm counts, divided by the total number of firms (including those that responded with "Don't Know" or missing). The shares are then scaled up by the total number of non-missing and "Don't Know" responses for each technology. The 2017 industry figures from the LBD are the figures listed in the tables. Industry tabulations for multi-unit firms are generated from the largest payroll industry within the firm (if there is a tie, then the industry with the most employment is used).

Table A2: Large CBSAs by AI Usage Rate among Single-Unit Startups

CBSA	(1) Firm-Weighted Share	Rank	(2) Emp.-Weighted Share	Rank
Atlanta-Sandy Springs-Roswell, GA	6.1	18	12.2	7
Austin-Round Rock, TX	5.4	33	15	5
Baltimore-Columbia-Towson, MD	3	49	6.3	28
Boston-Cambridge-Newton, MA-NH	6.7	13	7.6	19
Charlotte-Concord-Gastonia, NC-SC	5.6	29	7	23
Chicago-Naperville-Elgin, IL-IN-WI	5.4	33	10.3	9
Cincinnati, OH-KY-IN	5	39	5.4	33
Cleveland-Elyria, OH	4.5	40	6.9	25
Columbus, OH	4.5	40	16.1	4
Dallas-Fort Worth-Arlington, TX	6.8	10	6	30
Denver-Aurora-Lakewood, CO	4.5	40	2.4	48
Detroit-Warren-Dearborn, MI	5.7	25	8.1	16
Hartford-West Hartford-East Hartford, CT	6.8	10	2.5	47
Houston-The Woodlands-Sugar Land, TX	6.2	17	8.2	15
Indianapolis-Carmel-Anderson, IN	3.8	45	2.9	45
Jacksonville, FL	4.5	40	3.3	41
Kansas City, MO-KS	6.6	15	6.1	29
Las Vegas-Henderson-Paradise, NV	7.7	4	3.2	43
Los Angeles-Long Beach-Anaheim, CA	5.8	23	9.4	10
Louisville/Jefferson County, KY-IN	5.6	29	18.8	3
Miami-Fort Lauderdale-West Palm Beach, FL	5.8	23	4.2	36
Milwaukee-Waukesha-West Allis, WI	4.2	44	8.3	14
Minneapolis-St. Paul-Bloomington, MN-WI	5.7	25	7	23
Nashville-Davidson–Murfreesboro–Franklin, TN	8.3	2	11.8	8
New Orleans-Metairie, LA	7.4	5	4	37
New York-Newark-Jersey City, NY-NJ-PA	5.5	32	8.6	13
Oklahoma City, OK	3.8	45	5.8	31
Orlando-Kissimmee-Sanford, FL	5.1	37	2.3	49
Philadelphia-Camden-Wilmington, PA-NJ-DE-MD	6	19	7.5	20
Phoenix-Mesa-Scottsdale, AZ	5.9	21	7.8	17
Pittsburgh, PA	5.1	37	7.8	17
Portland-Vancouver-Hillsboro, OR-WA	5.4	33	6.9	25
Providence-Warwick, RI-MA	3.8	45	3.8	38
Raleigh, NC	6.9	9	3.3	41
Richmond, VA	5.6	29	4.3	35
Riverside-San Bernardino-Ontario, CA	5.9	21	20	1
Sacramento–Roseville–Arden-Arcade, CA	6.4	16	9.1	11
St. Louis, MO-IL	5.7	25	3.6	40
Salt Lake City, UT	5.7	25	5.7	32
San Antonio-New Braunfels, TX	8.3	2	3.8	38
San Diego-Carlsbad, CA	7.4	5	5.1	34
San Francisco-Oakland-Hayward, CA	9.5	1	18.9	2
San Jose-Sunnyvale-Santa Clara, CA	7.1	8	13.6	6
Seattle-Tacoma-Bellevue, WA	6.8	10	8.7	12
Tampa-St. Petersburg-Clearwater, FL	7.4	5	7.4	21
Tucson, AZ	6.7	13	6.4	27
Tulsa, OK	6	19	3.2	43
Virginia Beach-Norfolk-Newport News, VA-NC	3.6	48	2.6	46
Washington-Arlington-Alexandria, DC-VA-MD-WV	5.2	36	7.4	21
Mean Micropolitan (< 50,000 persons)	**4.1**	-	**8.2**	-
Mean Small-sized CBSA (50,000 - 250,000 persons)	**4.9**	-	**6.7**	-
Mean Medium-sized CBSA (250,000 - 1,000,000 persons)	**4.5**	-	**6.8**	-
Mean Large-sized CBSA (> 1,000,000 persons)	**5.9**	-	**7.7**	-

Notes: This table reports the AI usage rates among single-unit startups in large CBSAs. Large CBSAs are those with more than 1,000,000 persons. Usage rates are computed using a modified firm weight based on the national share of single-units within strata defined by size class, age class, and 2-digit NAICS sector. Unlike our "startup sample" described in Table 1 (and used for regressions in Tables 6a, 6b, and 7), the underlying sample of this table is not restricted to firms with owner or revenue information. Rather, the underlying sample is a subset of our baseline sample—specifically, single-unit startups located in large CBSAs (roughly 30,000 firms).

Table A3: AI Usage Rates by Firm Characteristics (%)
Startup Sample

Full Sample	AI Usage Rate (%) 6.0		-
	Yes	**No**	**Difference**
Primary Owner Characteristics			
Hold Advanced Degree	8.4	5.3	3.1
Owned Prior Business	6.4	5.8	0.6
Missing Age	7.7	5.9	1.8
Owner Age (0–34)	5.9	6	-0.1
Owner Age (35–54)	6.1	5.9	0.2
Owner Age (55+)	5.6	6.2	-0.6
"Very Important" Reasons for Owning the Business			
Wanted to be own boss	5.9	6.2	-0.3
Flexible Hours	5.8	6.2	-0.4
Balance work and family	5.9	6.2	-0.3
Opportunity for greater income	6	6	0
Best avenue for ideas/goods/services	6.5	5.3	1.2
Unable to find employment	7.5	5.9	1.6
Working for someone else not appealing	6.2	5.9	0.3
Always wanted to start own business	6.1	5.9	0.2
Entrepreneurial role model	6.6	5.8	0.8
Carry on family business	6.9	5.9	1
Help or become involved in community	7.6	5.5	2.1
Other reasons	8	5.8	2.2
Startup Conditions			
Funded by VC	18.8	5.9	12.9
Missing Startup Capitalization	4.5	6.2	-1.7
Startup Capitalization <25K	6	6	0
Startup Capitalization 25K–1M	6.6	5.6	1
Startup Capitalization 1M+	9.2	5.9	3.3
Don't Know Startup Capitalization	5.3	6.1	-0.8
Innovative Strategies & Outcomes			
Process Innovation	11.9	4.6	7.3
Product Innovation	7.4	4.4	3
Patents Owned or Pending	14.8	5.8	9
IP is very important	12.5	4.4	8.1
Growth-oriented innovation strategy	6.7	4.4	2.3
Geography			
In Micropolitan or Rural CBSA ($< 50,000$)	5.3	6.2	-0.9
In Small-Sized CBSA ($< 250,000$)	5.3	6.1	-0.8
In Medium-Sized CBSA ($< 1,000,000$)	5.8	6.1	-0.3
In Large-Sized CBSA ($1,000,000+$)	6.6	5.4	1.2

Notes: "Use" is defined as having responded with "In use for less than 5% of production or service", "In use for between 5%–25% of production or service" or "In use for more than 25% of production or service" for the category listed on any of the AI-based Business Technologies. The "Growth-oriented innovation strategy" indicator is equal to 1 if a firm responded that a focus on introducing new goods or services, reaching new customer groups, or opening up new domestic or export markets was a "very important" strategy for the business. The underlying sample for this table is our startup sample (column 4, Table 1).

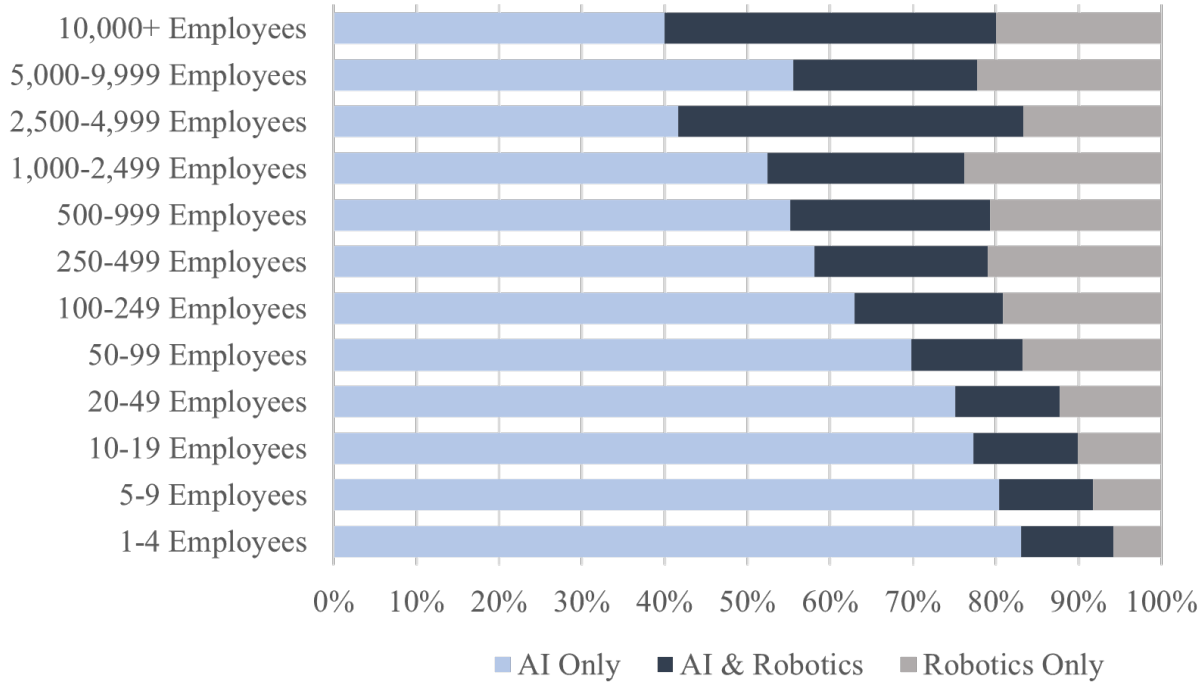
Table A4: Correlates of AI Adoption – including Early and Later Firm Growth Rates Owner & Revenue Sample

Description	(1) Use AI	(2) Use AI	(3) Use AI	(4) Use AI	(5) Use AI
Revenue Growth (First 3 years)	0.0075***	0.0073***	0.0061***	0.0074***	0.0021
	(0.0016)	(0.0016)	(0.0016)	(0.0016)	(0.0016)
Revenue Growth (Last 3 years)	0.0093***	0.0088***	0.0083***	0.0093***	0.0041**
	(0.0019)	(0.0019)	(0.0019)	(0.0019)	(0.0018)
Advanced Degree (1/0)		0.0148***			0.0087***
		(0.0019)			(0.0018)
Prior Business (1/0)		0.0102***			0.0050***
		(0.0013)			(0.0013)
Owner Age (35-54)		-0.0039			0.0004
		(0.0026)			(0.0026)
Owner Age (55+)		-0.0082***			-0.0014
		(0.0027)			(0.0027)
Lifestyle Reason (1/0)		0.0012			-0.0021
		(0.0013)			(0.0013)
Funded by Venture Capital (1/0)			0.0770***		0.0578***
			(0.0103)		(0.0101)
Startup Capitalization 25K-1M (1/0)			0.0159***		0.0090***
			(0.0015)		(0.0015)
Startup Capitalization 1M+ (1/0)			0.0271***		0.0130***
			(0.0044)		(0.0044)
Process Innovation (1/0)					0.0574***
					(0.0020)
Product Innovation (1/0)					0.0090***
					(0.0012)
Patents Owned or Pending (1/0)					0.0329***
					(0.0064)
IP is very important (1/0)					0.0572***
					(0.0021)
Growth-oriented innovation strategy (1/0)					0.0091***
					(0.0012)
Small CBSA				0.0012	0.0003
				(0.0023)	(0.0022)
Medium CBSA				0.0017	-0.0004
				(0.0019)	(0.0019)
Large CBSA				0.0046***	0.0013
				(0.0017)	(0.0017)
State-by-Ind Controls	Yes	Yes	Yes	Yes	Yes
Firm Age Controls	Yes	Yes	Yes	Yes	Yes
Owner Gender Controls	Yes	Yes	Yes	Yes	Yes
Observations (rounded)	209,000	209,000	209,000	209,000	209,000
R-squared	0.125	0.126	0.127	0.125	0.148

Notes: Standard errors in parentheses. *, ** and *** denotes significance at the 10, 5 and 1% respectively. Revenue growth refers to the three-year average of the log-difference measure of annual revenue growth. "Use" is defined as having responded with "In use for less than 5% of production or service", "In use for between 5%–25% of production or service" or "In use for more than 25% of production or service" for the category listed on any of the AI-based Business Technologies (Automated Guided Vehicles, Natural Language Processing, Machine Learning, Machine Vision and Voice Recognition). State-by-Ind controls refers to state and 6-digit NAICS industry dummies. The underlying sample for this table is our owner and revenue sample (column 3, Table 1). Note that Column 5 is identical to column 7 of Table 7.

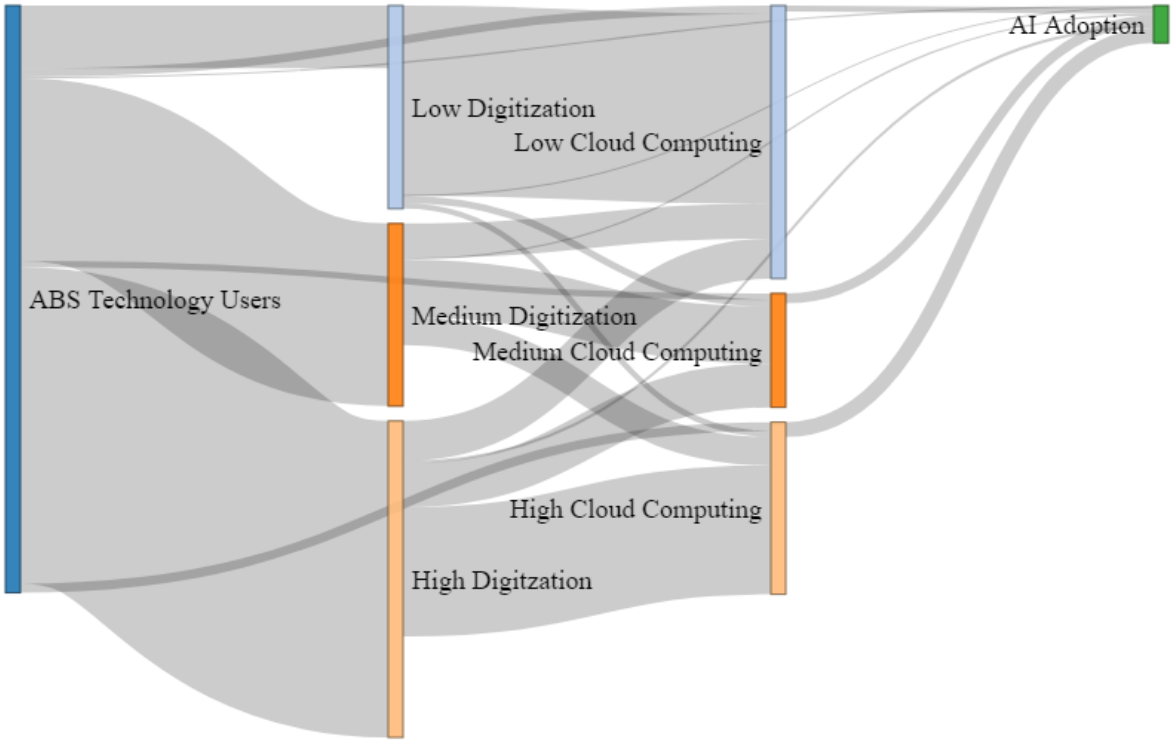
Appendix Figures

Figure A1: AI and Robotics Use by Firm Size
Full ABS Sample



Notes: This figure reports the percentage of firms that use AI or robotics among firms in the ABS–LBD linked sample (column 1, Table 1) that reported using either an AI-based technology or robotics (roughly 34,000 firms). AI use is defined by the weighted share of firms by sector that indicate intensity of use of at least 1 of the following business technologies: Automated Guided Vehicles, Machine Learning, Machine Vision, Natural Language Processing and Voice Recognition. Robotics use is collected from the same survey question (see Table 2) and similarly weighted.

Figure A2: Technological Interdependencies by Intensity



Notes: Sankey representation of firm counts in the ABS–LBD linked sample (column 1, Table 1) as they progress from no technology adoption to reliance on digital information, then cloud computing, and finally AI adoption. The size of the grey area is representative of the number of firm progressing to the next "stage" of technology use. Note that the calculations are made using imputed values for missing responses. Use of AI constitutes the weighted count of firms that indicate use intensity of at least 1 of the following business technologies: Automated Guided Vehicles, Machine Learning, Machine Vision, Natural Language Processing and Voice Recognition (Table 2). Low, Medium, and High intensity use of Digitization and Cloud Computing refer to the respondent answering "All" (corresponds to "High"), "More than 50%" (corresponds to "Medium") and "Less than 50%" corresponds to "Low" for at least one Digital Share of Business Activity or Cloud Service Purchases.

B Data Appendix

B.1 Overview of the ABS

The Annual Business Survey (ABS) was first conducted in 2018 in partnership between the Census Bureau and the National Center for Science and Engineering Statistics (NCSES) within the National Science Foundation. It covers all non-farm employer businesses filing the IRS 941, 944, or 1120 tax forms (which includes publicly traded firms). Firms surveyed are selected based on a sampling method that stratifies firms by state, industry, and ownership ethnicity and gender (for details on the 2018 ABS sampling methodology, visit https://www.census.gov/programs-surveys/abs/technical-documentation/methodology.2018.html).

The ABS consolidates three earlier data collections: the Survey of Business Owners (SBO), the Annual Survey of Entrepreneurs (ASE), and the Business R&D and Innovation Survey for Microbusinesses (BRDI-M). The SBO was conducted by the Census Bureau; the ASE was conducted by the Census Bureau in partnership with the Ewing Marion Kauffman Foundation and the Minority Business Development Agency; and the BRDI-M was conducted by the Census Bureau under a partnership with NCSES. The ABS also includes an innovation section—in part derived from the 2016 Business R&D and Innovation Survey (BRDIS)—to provide an expanded set of nationally representative business innovation statistics As a result of these consolidations from earlier collections, the ABS captures many important features of U.S. businesses, including ownership and owner characteristics, startup or acquisition funding, initial capitalization and firm financing, intellectual property strategy, and several aspects of innovation and R&D. The only purpose-designed questions for this inaugural wave of the ABS pertained to digitization, cloud computing, and advanced business technology use, as described in Section 3. See Zolas et al. (2020) for additional details on this technology module, and see https://www.census.gov/programs-surveys/abs/about.html for more details about the ABS overall.

B.2 Measuring Owner Characteristics, Startup Financing, and Firm Innovation and Business Strategies in the ABS

B.2.1 Education

The ABS asks "Prior to establishing, purchasing, or acquiring this business, what was the highest degree or level of school" the owner completed? There are 9 checkbox responses ranging from "Less than high school/secondary school graduate" to "Professional Degree, beyond a Bachelor's Degree (for example, MD, DDS, DVM, LLB, JD)." We construct an indicator of having an advanced degree, defined as having a Master's, Doctoral, or Professional degree.

B.2.2 Serial Entrepreneurship

The ABS asks, "Prior to establishing, purchasing, or acquiring this business, how many previous businesses has the owner owned?" Six possible responses range from 0 through 5 or more, though the important variation is captured by an indicator for any prior ownership.

B.2.3 Owner Age

Six checkboxes solicit owner age: under 25, 25–34, 35–44, 45–54, 55–64, and 65 or over. We collapse this into three informative categories: less than 35 years (omitted group), 35–54 years, and 55+ years.

B.2.4 Owner Aspirations

The ABS asks about how important (very important, somewhat important, and not at all important) are twelve potential reasons for owning this business. These reasons include: "Wanted to

be my own boss," "Flexible hours," "Balance work and family," "Opportunity for greater income," "Best avenue for my ideas/goods/services," "Unable to find employment," "Working for someone else didn't appeal to me," "Always wanted to start my own business," "An entrepreneurial friend or family member was my role model," "Wanted to carry on the family business," wanted to "Help and/or become more involved in my community," and "Other (Specify)." We construct an indicator which receives a value of 1 if the primary owner reports either "flexible hours" or "balance work and family" as a "very important" reason for owning the business.

B.2.5 Funding Sources

For sources of capital, there are twelve checkbox options provided and respondents can check all that apply. Options include personal/family sources (separately for savings, other assets, home equity loans, and credit cards), business credit cards, business loans (separately for government-guaranteed, bank or financial institution, and government), investment (separately for from family/friends and venture capital), grants, and other sources.

B.2.6 Funding Levels

Regarding the total amount of capital, ten checkboxes range from less than 5,000 dollars to 3 million dollars or more, including a "Don't Know" option: less than $5,000; $5,000–$9,999; $10,000–$24,999; $25,000–$49,999; $50,000–$99,999; $100,000–$249,999; $250,000–$999,999; $1,000,000–$2,999,999; and $3,000,000 or more. We collapse these ranges into three broader ranges: less than $25,000; $25,000–$999,999; and $1,000,000 or more.

B.2.7 Innovation

The ABS asks about process and product innovation, separately, across the prior 3 years (2015–2017). The process innovation question asks whether the business introduced new or significantly improved manufacturing methods; logistics, delivery, or distribution methods; and supporting activities for such processes (such as maintenance systems or operations for purchasing, accounting or computing). The product innovation question asks whether the business introduced new or significantly improved goods or services. Response options to both are "yes," "no," and "not applicable." We construct indicators for each.

To measure the firm's reliance on formal IP, we first combine two questions that capture the number of U.S. patent applications pending and the number of U.S. patents owned by the business by the end of 2017, collapsing them into an indicator of any patents owned or pending.

We also leverage questions on the ABS about the level of importance (very important, somewhat important, and not at all important) of the following six types of intellectual property protection: utility patents (patents for inventions), design patents (patents for appearance), trademarks, copyrights, trade secrets, and nondisclosure agreements.

B.2.8 Business Strategy

In the section on Innovation, the ABS elicits the importance (very important, somewhat important, and not at all important) of fourteen business strategies over 2015–2017. These strategies include: improving existing goods/services, introducing new goods/services, reaching new customer groups, customer-specific solutions, lowering price, reducing costs, satisfying key customers, developing niche/specialized markets, opening up new domestic markets, opening up new export markets, improving internal processes, improving delivery of existing goods/services, improving workforce, and understanding/meeting customer needs. We consider a business as having a "growth-oriented" strategy if it reported that introducing new goods or services, reaching new customer groups, or opening up new domestic or export markets were "very important."

B.3 Tabulation Weights for Geographic Results

Because the samples for Figures 5a, 5b, 6a, 6b, and Table A2 contain only single-unit firms, these differ somewhat from our nationally representative baseline sample described in Section 3.1. We thus modify our firm weights. Instead of generating weights based on the universe of firms in the LBD, we generate weights to represent the universe of single-unit firms in the LBD. The stratification we use for calculating the new weights does not explicitly include geographic information. A more complete dive into the geography of AI diffusion would likely benefit from a more nuanced weighting scheme that directly incorporates geographic information in some manner. Regarding the method of generating weights, our non-parametric weighting scheme ensures that the weighted number of surveyed ABS firms is equal to the number of firms in the same respective strata in the LBD. But this relies on a critical mass of surveyed firms in each stratum to avoid assigning unusually large values as weights, and including geography as an additional stratum would result in very small cells. We thus recommend a parametric weighting approach for researchers aiming to incorporate geography directly into the design of firm weights.